

GoodRegressor

User manual and worked output guides

GoodParser | GoodDesigner | GoodCurator | GoodRegressor

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This manual describes GoodParser, GoodDesigner, GoodCurator and GoodRegressor, together with the worked examples Example001_BCCDuctile and Example002_SocialPsychology. The algorithm is introduced in Seong-Hoon Jang, *GoodRegressor: A Hierarchical Inductive Bias for Navigating High-Dimensional Compositional Space*, doi:10.48550/arXiv.2510.18325, including its Supplementary Information (SI).

The emphasis is practical: what each input controls, how files connect, what the example actually produced, and which conclusions the outputs do and do not support. Values labelled **Example** below are values in the supplied files, not universal defaults or recommended settings for every dataset.

Essential terminology. In GoodRegressor.cpp and its model reports, the legacy label test means **validation**: these data participate in model selection. It does not mean an independent held-out test set. Literal filenames, input keys and output headers are retained in this manual so that they remain searchable. The preprint's separately reserved outer benchmark test folds retain their genuine test-set meaning.

Scope. Chapter 1 explains the input files and parameters. Chapters 2 and 3 explain the saved outputs of the materials and social-psychology examples. The appendix provides an output-file dictionary, a source map and an explanation of the acceleration mechanisms. Example results are reported from the supplied calculations.

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Reader's map

Chapter 1 is the input reference. Read 1.1 before editing any configuration, then use the section for the module being run. Section 1.6 explains the small model-chain files, stacking coefficients and other auxiliary inputs that are easy to overlook.

Chapter 2 follows Example001 from raw formulae to the Yield and Fracture regression ensembles. It explains record counts, stage selection, fitted coefficients, model performance, interaction summaries and symbolic exports.

Chapter 3 follows Example002 from its numeric social-psychology records through cohort-specific model construction, interpretation, genuine outer-fold evaluation and the separate machine-learning benchmark. It explains why legacy validation results, full-cohort fit and coverage-filtered outer-test metrics must remain distinct.

Appendix A is an output-file dictionary. Appendix B maps the main operations to their source locations. Appendix C explains Cholesky updates, on-the-fly learning and active MPI, and is linked from the TargetCPUTimeBeforeSwap parameter. CodeOnly/Utility contains the selected-model and benchmarking scripts; input_examples accompanies this manual with the complete configuration fragments introduced below.

Reading sources and examples

Source references use the following short keys. Line numbers refer to the extracted source snapshot; they will move when the program is edited. Function names are provided where helpful.

Key	Source
p	GoodRegressor preprint, doi:10.48550/arXiv.2510.18325; main-text page numbers and printed SI page numbers
GP	CodeOnly/GoodParser.cpp
GD	CodeOnly/GoodDesigner.cpp
GC	CodeOnly/GoodCurator.cpp
GR	CodeOnly/GoodRegressor.cpp
E	A path relative to Example001_BCCDuctile/
E2	A path relative to Example002_SocialPsychology/
BM	CodeOnly/Utility/benchmark_v14.py and benchmark_pp_v14.py; the saved example calculations also include copies under E2, 005_MLBM.
ED	Wolfram Language, ElementData, used in preparing the general atomic-property tables.
SP	Related social-psychology preprint, doi:10.31234/osf.io/waqjn_v2.

The manual's workflow is **Parser -> Designer (descriptors) -> Curator -> Regressor -> Designer (analysis/prediction/design)**. A user with an already prepared numeric TSV can enter directly at Curator or Regressor. The parser and materials-specific descriptor machinery are not prerequisites for a general numerical regression problem. [P, SI S3-S9; GR, input reader.]

1. Input parameter reference

1.1 Shared conventions, compilation and execution

Configuration files are ordered records

Each executable reads its own configuration from the current working directory:

Executable	Required configuration filename	Parallel interface in this snapshot
GoodParser.x	basic_properties_GoodParser.txt	Serial C++
GoodDesigner.x	basic_properties_GoodDesigner.txt	MPI C++
GoodCurator.x	basic_properties_GoodCurator.txt	Serial C++
GoodRegressor.x	basic_properties_GoodRegressor.txt	MPI C++

The long text before the value on a configuration line is generally a descriptive label consumed by an ordered reader. It is **not** a free-form key/value configuration language. Keep the original order and required lines. Do not insert a comment or blank line among active fixed-position records. Whitespace separates tokens; TSV data tables use tabs. Sentinels such as STOP, Functypes and BetaRegression have control-flow significance. [GP 217-265; GC 103-234; GR 1290-1563; GD 3430-4050.]

Use a period as the decimal separator, whitespace-free file paths and numeric labels, and a final newline. Use LF line endings for Linux execution. Keep each fixed-position configuration record on its own physical line.

One working directory per independent job. The sources create temporary files and contain wildcard cleanup operations. Examples include Curator's *.temporary.txt, Regressor's *.reg, *.pr.txt and smt, and Designer's design-worker files. Never run multiple jobs in one directory or use a directory containing unrelated files matching these patterns. Work from a copy, not the only copy of archived results. [GC/GR/GD, cleanup branches.]

Build commands and dependencies

The supplied compile_example.txt gives these commands. Run them inside CodeOnly, preserving the bundled Eigen directory structure:

```
g++ GoodParser.cpp -o GoodParser.x -std=c++11
mpicxx GoodDesigner.cpp -o GoodDesigner.x -std=c++11
g++ GoodCurator.cpp -o GoodCurator.x -std=c++11
mpicxx GoodRegressor.cpp -o GoodRegressor.x -std=c++11
```

Eigen is the external linear-algebra library identified by the package. The MPI modules additionally require a working MPI compiler wrapper and runtime.

Executables read configuration/data relative to their working directory. For example, in a prepared, isolated Parser folder:

```
/path/to/CodeOnly/GoodParser.x
```

In an allocated MPI environment, a generic launch has the following shape; replace the process count and executable path with values appropriate to the allocation:

```
mpirun -np 8 /path/to/CodeOnly/GoodRegressor.x
```

This is a launch pattern, not a scheduler submission script. The preprint's large-core benchmarks are not minimum hardware requirements and do not establish the runtime of Example001 on a workstation. A configured 1000-second sampling stage is also not a 1000-second wall-time guarantee for the entire pipeline. [P, pp. 24-27; E, Regressor configurations.]

Keep numerical meaning and display names separate

AveXensity denotes the average gravimetric density (mass per unit volume), as distinct from the MolDensity descriptor. Retain the literal spelling AveXensity in numeric headers and setX. A LaTeX display label does not rename a numeric column. Record physical units separately: changing units changes the arguments of nonlinear transforms and is not automatically compensated by the program.

Treat zero, missing data and the finite VInfinite sentinel as different concepts. The normal Designer statistics omit zero-valued atomic attributes; an absent tabular value must not be silently converted to a meaningful zero. Check all transformed inputs and predictions for finiteness. Domain restrictions of powers, logarithms and divisions still apply. [GD, get_mamssk, 2019-2093; GR, get_v, 900-940.]

1.2 GoodParser: formulae and aligned target values

Role. Convert chemical-composition strings into an explicit block representation and append accompanying target values. Parser prepares the material records consumed by Designer; it does not calculate the full descriptor matrix. [P, SI S6; GP 217-265, 328-454, 616-673.]

Fixed input records

Position / parameter	Example	Meaning and operational consequence
1. InputFilename	BCCYield.txt	Raw composition file. Each data record must be parseable by this program's grammar.
2. PostscriptFilename	BCCYield_postscript.txt	Text/target values appended to parsed compositions by row position , not by a join on identifiers.
3. LogFilename	BCCYield_log.txt	Parsing diagnostics. Read this before accepting the final record count.
4. OutputFilename	BCCYield_output.txt	Intermediate formula-parsing output and associated composition diagnostics.
5. FinalOutputFilename	BCCYield_dat.txt	Final material records, including the appended postscript/target data; the handoff to Designer.
Following block list	None, then Og, then STOP	Assign species to composition blocks; these blocks carry through to subsequent material processing.

The working BCC configuration is:

```
InputFilename BCCYield.txt
PostscriptFilename BCCYield_postscript.txt
LogFilename BCCYield_log.txt
OutputFilename BCCYield_output.txt
FinalOutputFilename BCCYield_dat.txt
=====AtomBlocks(IfNone, "None")=====
None
Og
STOP
```

Atom blocks and formula grammar

None provides the fallback block for species not assigned to a named block. The example places ordinary alloy species in the first block and Og in its own block. A block can contain a list of species. The reader also supports a * no-count marker for block-level aggregate diagnostics; that option is not used for Og in the archived example. It must not be confused with the multiplication symbol in Designer's user-defined descriptors. [GP, block reader and final-output writer.]

The source contains handling for numeric decimal stoichiometry, nested parentheses/brackets, composition mixtures and selected molar/atomic/weight-percentage notations. This is not a general symbolic chemistry engine: resolve algebraic variables such as x or 1-x before submitting a formula, and verify any less common syntax in the intermediate output. Never infer successful parsing solely from an executable exit code. [GP 328-454.]

Examples of mixture notation accepted by the parsing workflow include:

```
Mg85Al15 + 5 wt% Ni
0.2(CeO2) + 0.8(Fe2O3)
```

The first combines an Mg-Al composition with a weight-percentage Ni addition. The second uses numerical mixture coefficients for two parenthesized oxide compositions. Keep the stated weight or mixture notation in the input; examine the explicit species-content pairs in the Parser output to check the resulting interpretation.

The final representation is divided by literal || separators. It contains identifying/formula text, explicit species-content pairs grouped by the configured blocks, aggregate diagnostic fields and the appended postscript. Preserve the separators and block order. Downstream code does not expect this representation to be a simple two-column formula/target CSV.

What Og means in Example001

The raw string Al0.214Nb0.714Ta0.571Ti1V0.143Zr0.929Og298.15 encodes both alloy content and a condition value. Designer's descriptor declaration is Og U Og * 1 L= T M= T, so the Og amount is extracted as the feature displayed as T. It is a temperature carrier in this example, **not a proposed oganesson-containing alloy**. The numeric headers do not explicitly declare the units of T, Yield or Fracture; this manual does not infer units from their magnitudes. [E, 001_Parser/BCCYield raw input; 002_Designer/BCCYield configuration, line 44.]

Because the archived Parser block is Og, not a no-count Og * block, aggregate mass/content diagnostics include the pseudo-species. Do not read those totals as the physical mass or atomic fraction of the alloy. Descriptor construction later handles the carrier differently; neither representation licenses chemical substitution of that carrier during materials design.

Checks before continuing

Compare raw-formula, postscript, successful-output and downstream-input record counts. Inspect at least one simple alloy, one fractional composition and one mixed/block case. Check that each target remains associated with the intended composition after any failures or reordering. The appending procedure is positional; manually deleting a failed formula but not its matching postscript can shift all later targets.

1.3 GoodDesigner: descriptors, ensemble analysis and design controls

Role. Designer has two substantially different uses. With no model files, it calculates descriptors from materials, atomic attributes and optional structural data. With model chains, it evaluates symbolic models, constructs or loads stacking ensembles, reports interaction statistics, generates feature-scan tables and can optimize compositions. The same configuration has records for both uses, so unused design values still occupy their prescribed positions. [P, SI S6, S8-S9; GD 3430-4050.]

Files and data interpretation: records 1-6

Parameter	Value type / example	Explanation
AtomDatListFilename	Path; BCC_general_atomdat .txt	Atomic-property lookup table, prepared using Wolfram ElementData [ED]. Descriptor names must match its column names. Always retain the Valence column , including when it is not selected as a regression descriptor, because nominal-valence bookkeeping uses it. Keep the corresponding table with the trained models.
FaceFilename	Path or -	Optional face-dependent data; the labelled format requires AtomName and Face as its first two columns. Unused in Example001.

Parameter	Value type / example	Explanation
StructureListFilename / UseStrName	BCC_strdat.txt 0	Structural descriptor table and structure-name selection. The BCC example uses the None structure with 0. For multiple named structures, the supplied reader uses 1; see the format example below.
MaterialsFilename / HeadExists / PostTagNames	BCCYield_dat_shuffle d.txt 0 Yield	Parsed materials, whether a header is present, and names of trailing reference/target columns. These names also associate observations with model groups.
RegModelFilename	List of chain files, -, or list ending in !	-: descriptors without models. A list: load model chains. A standalone ! token on the list disables further composition design, allowing analysis/screening.
MapOutSpaceFilename	Path or -	Map-space input is parsed, but the associated main processing call is commented out in this source snapshot. Do not advertise this as a verified turnkey mode.

Designer uses MPI in the supplied code, even though the preprint's high-level workflow particularly emphasizes MPI for the Regressor. Use the build command for the actual source, not an inferred serial Designer build.

PostTagNames are literal identifiers. The group name Yield in a model-chain file must match the Yield post-tag when observations are needed for fitting/evaluation or offset correction. A file prefix such as Y005001.txt does not itself define the target group. Grouping is based on metadata inside the chain file. [GD, read_reg_chain and material/model matching.]

Structural data: StructureListFilename, S and B

The materials record carries its structure label separately from the parsed composition:

```
(DataEntryName) || (StrName) || (Parsed composition) || (Target metrics)
```

The parsed-composition field may itself contain || separators for its site/group blocks. Match the StrName in each materials record to the corresponding row of StructureListFilename when using multiple named structures. In the supplied reader, UseStrName = 1 enables that lookup; 0 uses the None structure. [GD, read_strdat; main setup near 9458-9462.]

The structure table begins with the descriptor list and the ordered site list. Its header then gives **descriptor-major, site-minor** columns: all sites for the first descriptor, then all sites for the next descriptor, and so on. Terminate the table with STOP.

```
DescriptorNames Coor OmegaA OmegaS PV
SiteNames A B C
```

For the BCC/FCC example, the complete header order is:

```
StrName Coor_A Coor_B Coor_C OmegaA_A OmegaA_B OmegaA_C OmegaS_A OmegaS_B OmegaS_C PV_A PV_B
PV_C
```

The following table displays the same two structure rows vertically so that every value remains readable. The ready-to-use row-oriented file is input_examples/structure_BCC_FCC.txt.

Column	BCC	FCC
Coor_A	14	12
Coor_B	X	X
Coor_C	X	X
OmegaA_A	0.866646249	0.897597901
OmegaA_B	X	X
OmegaA_C	X	X
OmegaS_A	0.248872795	0.399887002

Column	BCC	FCC
OmegaS_B	X	X
OmegaS_C	X	X
PV_A	53.05512343	54.90571602
PV_B	X	X
PV_C	X	X

X means that the specified site does not exist in that structure; it is not a numerical zero. Coord is the coordination number. OmegaA is the mean site-to-face solid angle of the polyhedron formed by surrounding atoms, OmegaS is the standard deviation of those solid angles, and PV is the polyhedron volume. Keep the units and geometrical definitions used to prepare the supplied values consistent throughout the input.

Declare these structural quantities in the AnalyzeDescriptors block as follows:

```
Coord S L= n_{c} M= Subscript["n", "c"]
OmegaA S L= \Omega_{m} M= Subscript["\Omega", "m"]
OmegaS S L= \Omega_{\sigma} M= Subscript["\Omega", "\Sigma"]
PV S L= V_{p} M= Subscript["V", "p"]
```

S means **StructureOnly**: the value is selected from the structural descriptor/site table.

For a descriptor that depends on both the species and its structural environment, use **B (Both)**. The following coordination table illustrates the site values used to select coordination-dependent ionic radii:

```
DescriptorNames Coord
SiteNames A B C
```

StrName	Coord_A	Coord_B	Coord_C
Perovskites	12	06	X
Perovskite-related,brownmillerites	07	05	X
Perovskite-related,Ruddlesden-Popper	09	06	X
Perovskite-related,Dion-Jacobson	08	05	06
Perovskite-related,hexagonal	12	06	X
Perovskite-related,others(V-layered)	05	06	X
Perovskite-related,others(Ti-layered)	06	06	X
Perovskite-related,others(Nd-layered)	06	07	06
Perovskite-related,others(A-site-vacancies)	08	06	X
Fluorites	08	X	X
d-Bi ₂ O ₃	08	X	X
BIMEVOX	05	06	X
Scheelites	08	04	X
Apatites	08	04	X
Cuspidines	07	04	X
Melilites	08	04	X
Garnets	08	05	X
Pyrochlores	08	06	X
Columbites	06	06	X
a-SrSiO ₃	08	04	X

StrName	Coor_A	Coor_B	Coor_C
Bi2UO6	05	10	X

The corresponding row-oriented input, including its header and final STOP, is input_examples/structure_coordination.txt. These are the supplied descriptor assignments for the example, rather than a general crystallographic definition of every structure in a class.

Then add the following declaration to AnalyzeDescriptors:

```
Radius B Coor
```

Designer uses the site's Coor value to select a property column from AtomDatListFilename: for example, 04 selects Radius04, 05 selects Radius05, and 06 selects Radius06. It then obtains that radius for the corresponding species. This is the mechanism for using coordination-dependent Shannon radii without assigning one radius to every environment. Preserve the zero-padded coordination strings and provide the matching Radius04, Radius05, etc. columns in the atomic table.

Objective, reliability and scale: records 7-11

TargetVReg supplies the desired value for each model group, in the target's working units. It is a target-location objective, not an implicit instruction to maximize every property. For the revised single-target Fracture setup, the desired value is 10000.

TargetStdDvReg provides a maximum permitted inter-model prediction dispersion for each group. Its units follow the corresponding modeled target, including any logarithmic transformation of that target. This is an ensemble-consistency screen, not a calibrated probability, confidence interval or error guarantee.

ZDistWeight supplies the per-target weights in the standardized distance to the desired target. Higher weight gives that target more influence in the distance. Weights belong to target groups, not to individual models in the ensemble. Use a nonnegative, deliberately chosen set for a distance objective.

UseAve and **UseStdDv** supply optional target-centering and scaling values; - leaves a value to be determined by the program. A user-supplied standard deviation must be a meaningful positive scale. Do not derive an evaluation scale from all-zero placeholders and then interpret it as a measured-data standardization.

Removing the standalone ! from the model list enables further materials design. For that use, provide TargetVReg, TargetStdDvReg, ZDistWeight, UseAve and UseStdDv with one entry for each target metric, in the same model-group order. For two target metrics, each record therefore needs two entries. Further multi-target composition design with this setup is not demonstrated in these examples.

Composition search: records 12-20

ModulateContentFactor is a list controlling allowed amount changes and doping proposals. Positive values are content multipliers, such as 0.9 1.0 1.1. Nonpositive entries are read through the doping-factor branch using their absolute values; the archived post-processing example uses 1.0 -0.1, corresponding to no positive scaling change plus a 0.1 doping fraction. Do not interpret -0.1 as a negative atom count. Avoid adding unexplained zero or negative factors merely to populate the list. [GD, parameter reader and material modulation branches.]

UseOffset is 0 for direct model predictions or 1 for an observed-value-anchored change. With 1, matching reference post-tags must be present. The latter prediction convention is:

```
y_modified = y_observed_original
              + f(modified composition) - f(original composition)
```

This is distinct from f(modified composition). State which convention is used when reporting candidates. It does not validate the extrapolated difference. [P, SI S42, Eq.S65; GD, offset branches.]

MaxIteration limits the composition-improvement iteration loop; Example001 uses 100. It is not the number of Regressor bagging runs, MPI ranks or depth stages.

NeighborSelectSpace lists atomic-attribute names used for replacement-neighbor selection, such as Electronegativity MetalRadius or the same list with MValEleN. These are atomic attributes, not already aggregated Ave... descriptor names.

NeighborZCrit has two operational forms. A positive first value selects a scalar standardized-distance criterion. A nonpositive first value selects componentwise bounds, using the magnitudes of the listed entries in the corresponding attribute dimensions. The example -0.3 -0.5 therefore means separate difference limits; it is not a negative Euclidean radius. Ensure the number and ordering match the selected neighbor attributes. [GD, neighbor-selection setup.]

ExcludeNeighbors removes specified species from the proposed replacement-neighbor list. It accepts explicit species or the predefined groups below. These are the program's literal lists, not general definitions of toxicity, rarity or electronic configuration.

Input group	Species excluded
Toxic / toxic	Hg Pb Cd As Tl Be Cr Cs Po At Rn Fr Ra Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr Rf Db Sg Bh Hs Mt Ds Rg Cn Nh Fl Mc Lv Ts Og
Rare / rare	Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
Non-quenched / NQ	Mn Fe Co Ni Cu Ru Rh Pd Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Os Ir Pt Au

Use NQ as the no-hyphen spelling for the last group in the supplied reader. A hyphen disables the ExcludeNeighbors record before that reader reaches its group-name matching. Explicit species lists can be used instead. [GD, exclusion-list reader, 3616-3715.]

DoNotFindNeighbors distinguishes protected species from partial-doping-only species. Unprefixed names are not substituted with other species. A name prefixed by - can remain unchanged or be partially substituted, but is not fully replaced. The migrating-ion example below illustrates both forms. Also inspect any <species>.nb.txt files supplying explicit neighbor lists.

AnionName selects the composition/charge bookkeeping. A single anion name uses that anion together with AnionValence; - disables that single-anion mechanism. The extended * form instead supplies species/nominal-valence pairs whose contents can adjust for charge compensation, as shown below.

AnionValence gives the charge in single-anion bookkeeping. Its numerical value is not used for the AnionName * ... example below, which supplies the valences directly. Retain the fixed-position record in the configuration even when its value is unused.

Example: protect migrating ions while permitting charge compensation

```
DoNotFindNeighbors(IfNone,"-") Li Na Mg Ag K Ca Zn -H -N -F -S -Cl -Se -Br -I -O
AnionName(IfNone,"-"/YouCanAddCationNamesAlso...Try) * Li 1 Na 1 Mg 2 Ag 1 K 1 Ca 2 Zn 2
limit 0.1
```

Keep each displayed record on one physical line in the input file; line wrapping on this page is for reading only. The same records are supplied in input_examples/migrating_ion_controls.txt.

In DoNotFindNeighbors, Li Na Mg Ag K Ca Zn are protected from substitution. The prefixed entries -H -N -F -S -Cl -Se -Br -I -O allow only partial substitution (doping) or no change.

The leading * in AnionName activates the species/nominal-valence list. When other atoms are doped or replaced with different nominal valences, the amounts of the listed species can be modulated for charge compensation. Their identities remain protected even though their contents can change. In this branch, limit 0.1 bounds the candidate content relative to its starting content within factors 0.9 and 1.1. AnionValence is unnecessary for this calculation because the valences are given in the list. **Always leave Valence in the atomic-property table** for the remaining nominal-valence bookkeeping. [GD, extended anion reader and content check near 7465.]

The oxide constraints described in the preprint are settings for those demonstrations, not universal guarantees of phase stability or experimental feasibility. Neither neighbor similarity nor small ensemble dispersion replaces structural or experimental validation. [P, SI S42-S43.]

Outputs, statistics and interaction analysis: records 21-27

Parameter	Example	Meaning and caution
VInfinite	1000000000000000000	Finite sentinel used in exceptional/undefined numerical situations. Do not treat it as a physical measurement or as an accuracy setting.
Verbose	0	Diagnostic verbosity; 1 requests more detail.
OutputFilename	BCCYield_designeroutput.txt	Main report, including model/stacking information and design progress when enabled.
AnalysisOutputFilename	BCCYield_anaoutput.txt	Descriptor/analysis table. An optional integer on the same record supplies the analysis mesh.
DoNotAnalyze	-	Species excluded from descriptor statistics. This is not a list of descriptor columns to remove.
AnalyzeInteractionLevel	2 3	Maximum interaction-set level and optional number of chains. Use 1 1 when detailed descriptor/interaction analysis is not needed: that analysis can become a calculation bottleneck.
SkipMinMaxKurtoInAnalyzeInteraction	1	Exclude Min/Max/Kurto from interaction analysis. 0 retains them. A distinct 2 branch is mean-only analysis and must not be treated as an ordinary Boolean. It retains zero-valued attributes in get_mamssk; Example002 uses it for singleton numeric observations.

The 1 setting does not remove Min/Max/Kurto columns from the descriptor table. Example001's initial analysis still contains all six statistics for each selected atomic attribute; Curator later chooses the intended modeling subset. [GD, analysis/output branches; E, 002_Designer tables.]

Descriptor declaration block

After the fixed records, preserve the descriptor-block heading, add declaration rows, then terminate with STOP.

```
Mass A L= M M= M
ShearModulus A L= G M= G
Og U Og * 1 L= T M= T
STOP
```

A denotes an atomic attribute. S denotes a structural attribute. B denotes a value depending on both atomic and structural information and requires the corresponding postfix/mapping information. These modes must agree with the atom/structure tables. A specialized P branch reads an index and converts to atomic-style handling internally; it is not demonstrated here and is not documented as a general portable descriptor recipe. [GD, descriptor reader and post-processing of descriptor definitions.]

U defines a quantity from species contents. The numerator and denominator/product side may select species, use All, exclude species using !, or contain a fixed number. Exactly one * or / operator separates the two sides. For example, n U Na * 1 retains sodium content, while a selected-species sum divided by All defines a content ratio. The Og declaration extracts the stored condition without averaging it with alloy atom attributes. L= and M= are display/export strings for LaTeX and Mathematica; they do not change numerical evaluation.

More U declarations: contents, fractions and ratios

```
n U Na * 1
rTR U Ni Sc Ti V Cr Mn Fe Co Cu Zn Y Zr Nb Mo Tc Ru Rh Pd Ag Cd Hf Ta W Re Os Ir Pt Au Hg Rf
Db Sg Bh Hs / All
rTRe U Ni Sc Ti V Cr Mn Fe Co Cu Zn Y Zr Nb Mo Tc Ru Rh Pd Ag Cd Hf Ta W Re Os Ir Pt Au Hg Rf
Db Sg Bh Hs / ! Ni Sc Ti V Cr Mn Fe Co Cu Zn Y Zr Nb Mo Tc Ru Rh Pd Ag Cd Hf Ta W Re Os Ir Pt
Au Hg Rf Db Sg Bh Hs
```

n is the Na content multiplied by 1. rTR is the sum of the explicitly listed species contents divided by the total content (All). $rTRe$ uses the same numerator, but the denominator excludes that same species list using $!$, so it is the ratio of the listed content to the remaining content.

For **either side** of a U declaration, choose an explicit species list, All, an exclusion list beginning with $!$, or a fixed number. Use exactly one operator, either $*$ or $/$, between the sides. A zero denominator is not a usable ratio. The long declarations above remain single-line records in `input_examples/AnalyzeDescriptors_U.txt`; page wrapping is only typographical.

Atomic statistics: what the numbers actually mean

For an attribute a and retained species contents c_j , normal Designer processing uses stoichiometric weighting. In compact notation:

```
w_j = c_j / sum(c_j over species retained for this attribute)
Ave(a) = sum(w_j * a_j)
StdDv(a) = sqrt(sum(w_j * (a_j - Ave(a))^2))
Skew(a) = sum(w_j * (a_j - Ave(a))^3) / StdDv(a)^3
Kurto(a) = sum(w_j * (a_j - Ave(a))^4) / StdDv(a)^4
```

The normal branch **excludes zero-valued atomic attributes** from the retained set, as well as excluded species. Thus the normalization may differ between properties. A zero can behave as missing/unavailable atomic information rather than as a true zero contribution. The separate mean-only branch changes this behavior. For zero spread the implementation handles skewness and kurtosis specially; a sentinel kurtosis is not a physical heavy-tailed distribution. [GD, `get_mamssk`, 2019-2093.]

Example001 selects 15 attributes: Electronegativity, Mass, Valence, MValEleN, MetalRadius, Xensity, MolDensity, Mfilling, BulkModulus, ShearModulus, PoissonRatio, ThermalConductivity, ThermalExpansion, DebyeTemp and MMSusc. Normal processing yields 90 atomic-statistic columns, plus the user-defined Og feature. Valence and MValEleN are distinct supplied attributes; do not merge them merely because both relate to electrons.

1.4 GoodCurator: filtering and first-level interactions

Role. Read a numerical feature table, select target/feature spaces, optionally handle duplicated names or composition proximity, and construct an initial pool of products/ratios. Its output is a report containing a tabular block, not a pure ready-to-read TSV file from first line to last. [P, SI S7; GC 103-234 and `dataset/output` routines.]

Parameter reference

Position / parameter	Example	Explanation
1. DataFilenamePrefix	BCCYield_anaoutput	Input prefix; the program appends .txt. Supplying the extension twice points at the wrong path.
2. LogOutputFilename	BCCYield.txt	Main Curator report, including the curated DATASET block. Despite the label, it is not merely a short log.
3. OutputPrecision	20	Decimal output precision, not source-data or floating-point accuracy.
4. NameDuplicateHandling	-1	-1: leave duplicates; 0: retain minimum target; 1: maximum target; 2: modal/histogram-based handling with supplied minimum, maximum and increment.

Position / parameter	Example	Explanation
5. XSpace	Ave StdDv Skew Og	Header-substring selectors for the base features. Matching is not an exact-name-only list.
6. InteractXSpace	Ave Og	Header-substring selectors for features eligible for first-level products/ratios; - disables this additional space.
7. ParseData	-	Optional parsed-material file for composition-based handling; disabled in Example001.
8. ParseGroup	2	Number of groups inside the parsed-composition field: one plus its internal count. Not the number of targets or folds.
9. setY	Yield * *	Target-column name and lower/upper bounds; * means no corresponding bound.
10. SubFilterX	-	Optional feature-name/lower-bound/upper-bound triples applied as record filters.
Following sampling block	Heading, then STOP	Optional postfix/action/factor rows for composition-proximity selection flags.

The Curator needs unambiguous identifiers. Its name-column detection uses a Name substring. Feeding a Designer table that still contains both Name and CompositionName is unsafe; prepare one intended Name column and numeric columns. Example001 removes the three other metadata columns before Curator. [GC, input header selection; E, 002/003 tables.]

NameDuplicateHandling operates on repeated data names, not on independent measurements automatically grouped by chemical similarity. Confirm that the identity policy suits repeated experiments at different conditions. A maximum-target policy is a scientific curation choice, not a generic remedy for every duplicate.

In this source, a hyphen can disable the entire SubFilterX line. Therefore a negative numerical bound is not automatically safe to enter under the displayed generic syntax. Check the reader before using such a bound. A working-copy filtering step outside the program may be clearer than relying on this ambiguous branch; record that preprocessing explicitly. [GC 103-234.]

Products and ratios

Curator iterates over selected feature pairs in their existing order. The demonstrated pool contains a product and a forward ratio for a pair, not both possible ratio directions. Ratio eligibility requires fixed nonzero sign over the data for both participating variables in the implemented screening. A column with zeros or a sign change can therefore remove an expected ratio even when it was named in InteractXSpace. [P, SI S7; GC, interaction construction.]

In Example001, XSpace chooses 46 base features: 15 means, 15 standard deviations, 15 skewnesses and Og. InteractXSpace chooses 16 of these: 15 means plus Og. All pairs represented in this example produce the demonstrated product/ratio pool:

```
46 base features + 2 * C(16,2) interactions = 286 predictors
286 predictors + Name + target = 288 table columns
```

Those are measured properties of this example's feature pool, not an unconditional feature-count formula for a new dataset.

Optional composition-proximity sampling

ParseData, ParseGroup and the sampling block let Curator identify a preferred sample among compositionally close records, using either the minimum or maximum target value.

```
ParseData(*.txt,IfNone,"-") materials_dat.txt
ParseGroup 3
```

Here materials_dat.txt is the parsed MaterialsFilename used by GoodDesigner. Leave the intervening fixed records in their original positions. In the sampling block, use:

```
Postfix_SamplingAction(TowardsMin0;TowardsMax1)_SamplingFactor
U0.01 1 0.01
STOP
```

A materials record has the following structure:

```
(DataEntryName) || (StrName or user-defined name) || (Parsed composition) || (Target metrics)
```

ParseGroup counts the groups inside the parsed composition. It equals one plus the number of || separators within that composition field, excluding the separators for the leading metadata and trailing targets. Thus three parsed composition groups require ParseGroup = 3.

Curator normalizes the composition contents so that their sum is 1. It compares records in that normalized-composition space. When their distance is below SamplingFactor, the action chooses which target value to favor: **0 (TowardsMin) retains the lower value; 1 (TowardsMax) retains the higher value.**

In the example, U0.01 1 0.01 favors the maximum target for compositions closer than 0.01 in that normalized space. The output named by LogOutputFilename gains a column U0.01: 1 marks a retained record and 0 marks a record not retained by this selection. Additional sampling rows can create additional postfix columns. The output flags can then be applied as data filters, for example U0.01 1 1 in Regressor's SubFilterX. Producing the flags and filtering on them are separate operations. [GC, pick_data and sampling-indicator output.]

Handoff to Regressor

Extract the rectangular block beginning after the DATASET marker, including its header and only its material rows. Exclude the report preamble and trailing statistics. Retain literal feature names, including parentheses and operators. Save the result as input.txt, then set Regressor's setX to the chosen predictor names and setY to the target.

The companion tools/extract_curator_dataset.py performs this extraction conservatively and checks that each selected row has the expected number of tab-separated columns. It refuses to overwrite an existing output. It does not train models, change descriptors or repair malformed numeric entries.

1.5 GoodRegressor: search, depth and statistical selection

Role. Search explicit symbolic forms under a controlled hierarchy. The preprint describes run-through, swap, transit, pick and repeated ensemble construction. The code searches a sampled lexicographic space and refines selected models; a reported best model is not proof of exhaustive enumeration or a globally optimal equation. [P, pp. 24-27; GR, search/refinement routines.]

Data and output records

DataFilename identifies the numeric TSV input; the reader can accept a list of compatible input files. All included tables must have consistent predictor/target headers and numerical meaning. Name is the optional observation identifier. Avoid whitespace inside feature names and preserve exact expression-style headers.

OutputFilename is the report prefix, output in the example. The generated filenames additionally identify the target, depth stage and requested term-count mode. output_Yield_2L4l.txt is not a separate model-target called 2L4l; it is the second stage of the Yield search with the indicated four-term limit and interaction mode.

OutputPrecision, here 20, controls printed precision. Twenty printed digits do not imply twenty reliable digits in a measured target, an optimized coefficient or a cross-platform rerun.

tFTestMesh, here 1000, controls the numerical mesh used by the implemented t/F significance calculations. It is not the number of folds, permutations or candidate models. Changing it changes the numerical significance calculation rather than the train/validation split. [GR 1290-1563 and statistical routines.]

NVariableLimit: the depth schedule

The example uses:

```
NVariableLimit(IfNegative,Limit) -5 -4I -3I -2I -1I
```

The magnitude is the number of active additive terms at the stage. A negative sign requests a strict limit, rather than permitting the corresponding additional-variable behavior; it does not request negative variables. The suffix selects the pool used at subsequent stages:

Token form	Meaning
-5	Strict five-term stage using the original selected input pool.
-4i	Strict four-term stage using the preceding stage's interaction pool.
-4I	Strict four-term stage using that interaction pool together with the original input pool.

The first stage has no preceding-stage interaction pool to reuse. Preserve capitalization: i and I are not synonyms. Output L marks the limited mode; the suffix identifies the interaction-pool mode. [GR, schedule parsing and inter-stage dataset construction.]

As the additive term count decreases, retained transformed expressions and their products/ratios can make individual terms more deeply composed. Thus fewer additive terms do **not** necessarily mean a simpler expression tree. The preprint's interaction-depth axis refers to this construction, not merely a generic feature-count sweep. [P, pp. 25-26; SI S8.]

The schedule supplies stages, not independent ensemble repeats. The example's 10 model members live in 10 run directories outside this one configuration. Do not describe -5 -4I -3I -2I -1I as five-fold cross-validation or five-member bagging.

Significance, time and local-refinement controls

PickUpPLevel is the significance threshold, 0.05 in the example. The full-Fisher/full-frequentist criterion described in the preprint requires the overall F-test and coefficient/intercept t-tests to pass the selected threshold. This is a model-admission rule inside a search; it is not a causal test of the resulting descriptors or an independent post-selection discovery probability. Look for the report's full_Frequentist flag rather than assuming every provisional printed model satisfies it.

Verbose selects log detail (0 or 1). It does not change the conceptual distinction between training, validation and external testing.

TargetCPUTimeBeforeSwap controls the initial sampling budget and the refinement options. For most cases, start with one of the following two presets:

```
TargetCPUTimeBeforeSwap -100 -1 !!
TargetCPUTimeBeforeSwap -1000 -1 !!
```

These are alternatives: keep **one** TargetCPUTimeBeforeSwap record in a configuration. Use -100 -1 !! for a shorter initial exploration, or -1000 -1 !! for a larger initial sampling budget. Both use the accelerated swap option and on-the-fly selection. The three fields have the following meanings:

Field	Meaning
-100 or -1000	Magnitude: run-through time target in seconds. Negative sign: enable the useSuperFastSwap branch.
-1	No positive time cap is imposed on the after-swap refinement by this field.
!!	Enable useSomeShotMode, the on-the-fly selection mode; do not enable useExitMode.

For the punctuation alternatives, ! enables only useExitMode, !! only useSomeShotMode, and !!! both. These flags are Regressor options, not shell syntax, and have a different meaning from Designer's standalone !.

Appendix C: Calculation-speed features explains Cholesky factorization updates, on-the-fly learning and the LUHMANN-to-HEGEL active-MPI workflow.

The run-through budget is not a total job timeout: later swaps, transformations, depth stages, I/O and MPI coordination add work. More CPU resources or altered time settings can change which models are visited. Use the two presets above as starting choices and retain the full configuration with the results. [GR, time-option reader; Appendix C.]

R2Threshold, 0.05, is used in the additional-variable improvement logic, including a training-R2 improvement check and associated validation conditions. It is not a universal minimum accepted R2, and it should not be presented as the convergence threshold for every stage of swap/transit. Strict negative term limits may prevent the addition that this setting controls. [GR, approximately 4707-4728.]

MaxBFGSIterations, 10, limits the relevant beta-regression numerical optimization. It does not limit the lexicographic search to 10 models or generate 10 bagged members. Baseline OLS is the demonstrated Example001 mode.

Splitting, duplicates and robustness

TrainRatio, 0.8, controls the training fraction. The remaining rows form the validation subset, although report fields call them test. For Example001 the resulting sizes are 215/54 for Yield and 122/31 for Fracture. Split membership can change during the staged computation; inspect identifiers rather than inferring membership from the ratio alone.

DuplicateX, 1, is the multiplicity of predictor entries in the search machinery; 1 keeps one copy, whereas values above one create repeated entries available to the search. It is not a request to duplicate observations and is not the bagging count. Repeated or highly collinear transformed predictors require interpretation of the actual fitted design matrix and diagnostics.

NameDuplicateHandling uses the same displayed policy family as Curator: -1 retains repeated names; 0/1 retain smaller/larger targets; 2 uses the provided histogram settings. Check name identity and condition information before discarding repeated names. Example001 uses -1.

NTestRobustness, 100, controls the number of robustness resamplings. The current routine selects subsets from the combined training and validation predictions and evaluates the already fitted model. It does **not** fit 100 new models. It is neither 100-fold independent validation nor an external test set. [GR 7041-7147; P, SI S8.]

The ROBUSTNESS summary reports the minimum, mean, maximum and standard deviation of each metric over the resamples. The corrected implementation takes the square root of the corresponding mean squared deviations for MSE, RMSE, MAE and R2. These describe metric variability across resampled subsets, not the inter-model prediction dispersion reported by Designer.

Target and predictor selection

setY supplies target names with lower/upper filtering bounds; * leaves a bound unrestricted. Example: Yield * *. Bounds filter records; they are not the optimizer's desired target. Designer's TargetVReg is a different input.

setX explicitly lists the predictor headers used by the Regressor. Unlike Curator's substring selectors, preserve the actual column/expression names here. The example lists all 286 curated predictors. A feature table can have extra unused columns, but numerical columns cannot be renamed without updating this list and any downstream model interpretation.

SubFilterX supplies additional feature-name/lower-bound/upper-bound triples, or - for no filter. For example:

```
SubFilterX(IfNone, "-"; {Name/Min/Max:IfNoLimit,Min/Max=*}*NSubFilterX) AveMass 0 50 U0.01 1 1
```

This retains only data entries satisfying $0 \leq \text{AveMass} \leq 50$ and $U0.01 = 1$. The latter applies Curator's composition-proximity selection flag. The names must match columns in the supplied numeric table.

Functypes: scalar transformations

The fixed records are followed by a Functypes heading and transformation rows. For example:

```
Functypes
1 ^1 pow a= 0 b= 1
2 ^-1 pow a= 0 b= -1
3 log10 log a= 0 b= 10 c= 1
BetaRegression(IfNo,"none") none
```

This is a minimal **illustration**, not the Example001 catalogue. Each row contains an index/display label, a function family and ordered coefficients. Keep row order stable for archived model evaluation. The printed display label is not an independently interpreted mathematical expression: actual evaluation follows the function family and coefficients. [GR, transform reader; GD, reading original Regressor configuration.]

Family	Implemented mathematical form
pow	$(x + a)^b$
log	$[\ln(x + a) / \ln(b)]^c$
sin, cos, tan	$[\text{func}((b * (x + a))^c)]^d$
sinh, cosh, tanh	$[\text{func}((b * (x + a))^c)]^d$
exp	$[d^{(c * (x + a)^b)}]^e$
erf, erfc	$[\text{func}(b * (x + a))]^c$
abs	$\text{abs}((x + a)^b)$

The last form computes the power before the absolute value; it is not necessarily numerically equivalent to $\text{abs}(x+a)^b$ for fractional powers on negative inputs. Some explanatory comments in the configuration differ from the implementation; the table above follows get_v. The erfc branch calls the complementary error function. [GR 900-940.]

The BetaRegression line terminates this read. Example001 has 67 active scalar-transform rows before that boundary at configuration line 86. Additional trigonometric transform listings later in the file are inactive text, not part of that run's search. The preprint's 109-transform setting must not be copied into the description of these archived runs. Read the rows before that boundary to identify the active families.

The example uses numerical constants such as 0.333333 and 2.71828; they are the supplied numerical values, not exact symbolic $1/3$ or the exact mathematical constant e . Use the original coefficients when reproducing predictions.

BetaRegression

none leaves the demonstrated baseline regression mode. The source supports beta-link options including logit, probit, cloglog, cauchit and nloglog in the beta setting. They belong on the actual controlling record as required by the reader, not on a later standalone explanatory line. Beta regression has additional numerical and bounded-response considerations and is not exercised by these BCC archives. Do not describe it as interchangeable with the validated OLS route without a dedicated check. [P, SI S47; GR, beta setup and optimization.]

1.6 Auxiliary inputs that control post-processing

Selected-model files for GoodDesigner

Files such as 005001.txt, Y005001.txt and F005001.txt are small manifests, not datasets or single flat equations. They identify the target, beta mode, original Regressor configuration and chronological stage reports needed to evaluate the selected model.

These selected-model files are generated to be supplied to GoodDesigner as a post-process. Use CodeOnly/Utility/find_maxR2.py to read the search summaries and create the files for the recommended models. The worked examples show the resulting manifests and their use in RegModelFilename.

The archived manifests contain absolute paths beginning `/work/k0736/k073600/GoodRegressor/`. Before relocation, rebase both the configuration path and every listed stage-report path in a copy. Ensure each path exists, the target group name is preserved, and the original transform catalogue is still the one used to train the model. Blindly moving only the small manifest is insufficient.

Optional text after the main chain boundary supports additional post-processing conventions, but no such advanced configuration is needed for the demonstrated selected models. Preserve the known-good syntax instead of inventing an extra formula block.

Utility scripts

The CodeOnly/Utility directory contains the companion scripts:

Script	Purpose
<code>find_maxR2.py</code>	Read the Regressor recommendations and generate selected-model files for Designer post-processing.
<code>benchmark_v14.py</code>	Run the conventional machine-learning and symbolic-baseline benchmark workflow.
<code>benchmark_pp_v14.py</code>	Post-process the benchmark logs and prediction files into summary tables.

Example002 demonstrates the benchmark inputs and outputs. Review the script's path and dataset settings for the intended working directory before execution.

Stacking coefficient files: `.se.txt`

Designer forms a consensus prediction of the form:

```
f_stack(x) = m0 + sum_i(m_i * f_i(x))
```

It is a least-squares stacking fit, not necessarily an arithmetic average. Coefficients can be negative. The progressive rows in an ensemble coefficient file correspond to accumulation of model members in their loaded order. Do not reorder the chain list while keeping the same coefficient file. [P, pp. 8-9; SI S8-S9; GD, ensemble construction.]

The source checks for target-named coefficient files such as `Yield.se.txt` and `Fracture.se.txt`. When these are present, they supply the archived stacking coefficients. Without them, the relevant path can fit coefficients to the current reference post-tags. For an unlabeled candidate grid with zero placeholders this would not be a meaningful training operation: carry the coefficients from the labeled fitting dataset instead. The dual-target folder includes these saved files.

The current Designer dispersion is calculated from individual-model predictions using absolute stacking weights around the unweighted member mean:

```
mean_member = sum_i(f_i) / number_of_members
ModelStdDv = sqrt(sum_i(abs(m_i) * (f_i - mean_member)^2)
                / sum_i(abs(m_i)))
```

This describes the supplied numerical routine, not a generic confidence interval. It is not the residual RMSE of the consensus predictor and it does not include every source of predictive uncertainty. [GD, `calc_bag_chain`, 3209-3233.]

Optional fold.txt

The optional fold input can restrict which material rows are analyzed. After its heading the source reads `NFolds`, `TakeFold`, training/test-side selection and straight/rotating assignment choices, with an optional `TestModelStdDevLimit`. `TakeFold` uses a one-based fold number. In straight assignment, membership is determined from row position in contiguous blocks; in rotating assignment it is row index modulo the fold count. [GD, optional fold reader near 9378 and material-selection logic.]

A useful workflow is to fit the stack with **Train0Test1 = 0**, then evaluate it with **Train0Test1 = 1**. The first run prints its stacking intercept and coefficients in the final block of the training analysis output. Copy that block into a target-named coefficient file, such as `satisfaction.se.txt`, in the test directory. GoodDesigner

reads this file automatically and uses the fixed coefficients for the test rows instead of fitting a new stack to them. Preserve the model order. Example002 shows this operation explicitly in Section 3.10.

This file does not automatically train all outer-fold Regressor models or prevent data leakage. A genuine outer-test evaluation requires the fitted descriptors/choices, model search and stacking fit to exclude those test outcomes. Record the exact folds and model manifests used. Any dispersion-based filtering of test predictions must report its retained coverage, not only the accuracy of the accepted subset.

Explicit neighbor files and structural metadata

A <species>.nb.txt file can override the automatically assembled replacement list. Search the working directory for these inputs when a prohibited or unexpected replacement appears. Configuration-only reasoning is incomplete when auxiliary files change the neighbor set.

BCC_strdat.txt in this example provides a minimal None structure and zero-valued coordination entries with named sites; it is not an atomistic BCC structure file and does not contain relaxed atomic coordinates. The alloy class comes from the example's data selection, not from a crystallographic reconstruction performed by the pipeline.

2. Example001: BCC Yield and Fracture

2.1 What is in the archive

Example001_BCCDuctile contains two labeled modeling tasks and a later candidate-screening use of their two ensembles. It is not the oxygen-ion-conductor benchmark shown in the preprint. Use the preprint for the algorithm and concepts, but use this archive for its actual inputs, model choices and results.

```
Example001_BCCDuctile/
  001_Parser/
    BCCYield/          raw formulae -> parsed materials
    BCCFracture/
  002_Designer/
    BCCYield/          parsed materials -> descriptors
    BCCFracture/
  003_Curator/
    BCCYield/          selected descriptors -> interactions
    BCCFracture/
  004_Regressor/
    BCCYield/005001 ... 005010/    ten staged searches
    BCCFracture/005001 ... 005010/
  005_Designer/
    BCCYield/          Yield ensemble, interpretation, design
    BCCFracture/       Fracture ensemble, interpretation, design
    BCCYieldFracture_DualDesign/    two-ensemble candidate screening
```

Quantity	Yield	Fracture
Labeled material records	269	153
First Designer descriptor-table columns	96	96
Prepared table columns supplied to Curator	93	93
Curated predictor columns	286	286
Regressor input columns including Name/target	288	288
Training / validation records per stage	215 / 54	122 / 31
Archived run directories	10	10
Depth stages per run	5	5

These counts were read from the supplied files and their model reports. The table headers lack explicit physical units for the targets; all RMSE and MAE values below are in the **stored target units**. The manual deliberately avoids guessing that a raw number must be MPa, percent or kelvin.

2.2 Stage 001: read the parsed result before calculating features

The first Yield raw record and its target illustrate the separation of chemical composition, condition and response:

```
Raw:    Al0.214Nb0.714Ta0.571Ti1V0.143Zr0.9290g298.15
Target: 1965
```

The corresponding parsed record separates the alloy species from the Og block and appends the target. Read the explicit atom-content pairs, not only the display name, because later output names may use shortened decimal strings. Og298.15 remains a condition carrier rather than a physical species in the alloy interpretation.

BCCYield_output.txt is useful for checking parsing and block assignment. BCCYield_dat.txt is the material-record handoff. BCCYield_log.txt should be consulted for parsing failures; an empty failure section is not a substitute for checking row correspondence. The equivalent Fracture files contain 153 material records.

2.3 Stage 002: understand the descriptor table

The materials files used by Designer are named BCCYield_dat_shuffled.txt and BCCFracture_dat_shuffled.txt. Their ordering differs from the preceding unshuffled materials files. Preserve each record's identifier and response when shuffling; never shuffle formula and target columns separately.

The first Designer configuration has RegModelFilename ... -, so the primary purpose is descriptor generation. Its analysis table has four metadata columns, 90 atomic-statistic columns, the condition feature Og, and one target:

```
No.  Name  CompositionName  Strcture
      [six statistics for each of 15 atomic attributes]
      Og  Yield
```

Strcture is the literal header spelling. The six statistics are mean, standard deviation, skewness, minimum, maximum and kurtosis. Their literal column names include the prefixes Ave, StdDv, Skew, Min, Max and Kurto. They should not all be assumed equally suitable for regression simply because they were generated.

The handoff is prepared, not automatic

The tables in 003_Curator are not byte-identical copies of the entire 96-column Designer table. The archived preparation removes No., CompositionName and Strcture, leaving Name and 92 other columns: all 90 atomic statistics, Og and the target. It also reflects numerical formatting/rounding in the intermediate table. The feature selection to 46 base variables occurs later inside Curator.

This distinction matters when rebuilding the example: feeding all 96 columns directly to Curator can trigger ambiguous Name matching, while stripping every Min/Max/Kurto column at this handoff would be a different, additional preprocessing decision. A faithful reconstruction separates metadata removal, numerical formatting and Curator's own feature selection. [E, 002_Designer and 003_Curator inputs; GC, header matching.]

2.4 Stage 003: read the Curator report and export the dataset

The example selects Ave, StdDv, Skew and Og for the base pool and Ave plus Og for interactions. Consequently the regression input contains both an individual feature such as AveMass and expression-style columns such as (AveMass*Og) or (AveMass/Og).

Open 003_Curator/BCCYield/BCCYield.txt. The preamble reports processing information. The DATASET section contains the header and 269 data rows. The report then resumes statistical summaries. The full report has 335 lines; it is not a 334-observation TSV. The Fracture report has 219 lines, of which 153 are data rows in the dataset section.

The companion extractor can be run as follows from a location where the paths resolve:

```
python tools/extract_curator_dataset.py \
  Example001_BCCDuctile/003_Curator/BCCYield/BCCYield.txt \
  work_yield/input.txt --expected-rows 269 --expected-columns 288
```

The output directory must already exist; the tool does not overwrite a file. Check the reported row/column count before using the result. The archived Regressor input.txt remains the reference when comparing a historic run: a freshly extracted table can preserve more printed precision than another intermediate preparation.

Curator's ratio columns have directional names. (AveMass/Og) and (Og/AveMass) are different numeric predictors; the example does not include every reversed ratio automatically. This becomes important when tracing a fitted term back to its underlying attributes.

2.5 Stage 004: a run is a five-stage search, not five independent tests

Each target has directories 005001 through 005010. Each contains an input.txt, a basic_properties_GoodRegressor.txt, and reports for the five requested stages. Within a target, the archived input/configuration files are identical across the ten run directories; the model results differ.

The nominal progression is five terms, then four, three, two and one, with the interaction pool expanded between stages according to the I schedule. It does not guarantee monotonically improving predictions. It also does not guarantee identical held-out memberships between stages.

Yield run 005001: why stage 2 was retained

Stage / report suffix	Training R2	Validation R2	Robustness mean R2	Validation RMSE
1L5	0.76801	0.90224	0.79523	161.0996
2L4I - selected	0.84341	0.78071	0.83295	195.7809
3L3I	0.82845	0.79416	0.83162	176.7308
4L2I	0.77439	0.81967	0.78557	217.2629
5L1I	0.72558	0.68000	0.72554	274.5299

The small manifest retains stage 2 and its stage 1 dependency. Stage1 has the largest single validation R2 in this run, but stage 2 has the largest reported robustness mean R2. Therefore selecting the file with the largest test R2 would **not** reproduce the archived selected chain.

Do not read this table as a fixed-test-set learning curve. The stage reports can use different training/validation memberships, and the robustness statistic reuses fixed predictions over resampled subsets of the combined data. Its comparison is the archive's internal model-selection diagnostic, not an unbiased external accuracy estimate. [E, 004_Regressor/BCCYield/005001 reports and manifest; GR, robustness routine.]

Fracture run 005001

Stage / report suffix	Training R2	Validation R2	Robustness mean R2	Validation RMSE
1L5	0.62201	0.66364	0.60838	9.4639
2L4I - selected	0.66305	0.43746	0.62938	10.6561
3L3I	0.55512	0.59469	0.54616	10.1247
4L2I	0.51509	0.47700	0.51081	9.8969
5L1I	0.35822	0.30333	0.35055	13.0969

Here, too, the archived selected stage is not the one with the highest single validation R2. The Fracture task remains more weakly reproduced by these individual selected models than the Yield task in the corresponding stored target scales; this observation does not by itself identify its physical cause.

The ten selected chains

Run	Yield selected stage / terms	Fracture selected stage / terms
005001	2 / 4	2 / 4
005002	3 / 3	2 / 4
005003	2 / 4	1 / 5
005004	3 / 3	1 / 5
005005	2 / 4	1 / 5
005006	2 / 4	2 / 4
005007	2 / 4	2 / 4
005008	3 / 3	2 / 4
005009	2 / 4	1 / 5
005010	3 / 3	2 / 4

The selection is read from the supplied manifests. Do not assume that all ensemble members have the same term count or the same number of predecessor reports.

Split reproducibility: what was actually checked

For each target, the first-stage validation identifiers are identical across all ten archived runs. The later stages inspected have different memberships across the ten runs. The source uses its random-selection mechanism but exposes no seed field in the documented configuration. These facts support neither the statement that all splits are identical nor the statement that ten fully independent first-stage partitions were established.

For future reproducible ensembles, record random initialization, row order, process count, time/search settings and the actual split identifiers. The preprint's intended repeated train/validation procedure must be distinguished from the split evidence available in this particular archive. A time-limited parallel search can yield different models even when its initial data partition is the same.

2.6 How to read a Regressor model report

Use 004_Regressor/BCCYield/005001/output_Yield_2L4I.txt as the reference example. The report includes an initial input/data section, search progress, a final answer, fitted-term statistics, training predictions, the legacy TEST block, robustness summaries and the next-stage interaction information.

FINAL_ANSWER and training statistics

In this report the FINAL_ANSWER marker is at line 807. The ensuing training summary gives four model degrees of freedom, 210 error degrees of freedom, training R2 0.8434054967531893, RMSE 200.911693128815 and MAE 158.398188270382. The flag full_Frequentist is 1.

ModelDF describes the fitted model terms in this output; ErrorDF is the residual degree-of-freedom count used for inference. SStot, SSres, MSE, RMSE, MAE, det_r2, det_radj2, Fvalue and p_Fvalue are not interchangeable metrics. A printed p-value of zero should be read as a numerical/printing result at extreme significance, not as a mathematical proof of exactly zero probability.

Coefficient table: separate prediction from interpretation

The fitted model is an intercept plus coefficients times symbolic terms. The report includes the raw coefficient, a standardized coefficient, statistical uncertainty/significance information and collinearity diagnostics. Use the raw coefficient and exact transformed predictor to reproduce predictions; standardized coefficients are for comparison within their specified normalization, not drop-in replacements in the raw-unit equation.

The selected Yield example has these raw coefficients, rounded here for reading:

Term	Raw coefficient	Standardized coefficient
1	9175.32518184	0.87856535
2	-2209.48771472	-0.33665076
3	-27221.24202102	-0.33404909
4	-2526.45783891	-0.11786138
Intercept	6365.29441034	Not a feature-effect ranking

For example, term 3 is (AveMValEleN/AveThermalConductivity)^2 in the actual fitted report. The sign of its coefficient applies to that **whole composite term**, with the remaining model terms held fixed. It is not by itself an unconditional causal effect of valence-electron number or thermal conductivity.

The leading term also combines temperature, thermal expansion, density and thermal conductivity through nested transformations. Expanding it by intuition from an abbreviated label is unsafe; preserve the stage-chain definitions and original Functypes parameters. Shared base variables can appear in several terms with opposing contributions.

Training and validation prediction rows

dataset.size() gives 215 training rows. The TEST marker at line 1045 introduces 54 validation rows. The validation summary gives:

```
R2      = 0.7807093932299739
RMSE    = 195.780927504701
MAE     = 164.667892647044
```

A validation record such as 204_MoNbTaTiVWOg1273 has stored target 752.8666667 and prediction about 546.0820898. This illustrates why a model's overall R2 should not be substituted for a per-material reliability check. Keep the observation name, actual target and predicted value together when plotting residuals or parity diagrams.

Terminology to use in captions: "Validation predictions; the program's legacy output label is TEST." Do not label this panel "external test". The parameters and structure were selected using validation performance during the search. [P, pp. 24-26; user terminology clarification; GR selection logic.]

ROBUSTNESS: sample size, mean and standard deviation

The example's block uses 100 resamples of size 54 from the combined fitted prediction pool. Its mean R2 is 0.832947975252808. The standard deviation corresponding to this example's resampled R2 values is approximately 0.03565.

Quantity	Value / meaning
Number of resamples	100
Rows per resample	54
Mean R2	0.83294798
StDev of R2	Approximately 0.03565

This statistic samples already available prediction errors, including training-data predictions. It provides the internal robustness/selection summary. It is not a substitute for a completely excluded test fold or a newly acquired dataset. [GR, robustness routine; P, SI S8.]

INTERACTION and predecessor reports

The interaction section supplies expressions that can become candidate terms in the next stage. Deleting it, trimming the report to a printed coefficient table, or replacing the original transform list can break reconstruction of a later chain. Archive reports and configurations as model dependencies, not merely human-readable logs.

2.7 Stage 005: build and inspect the stacking ensemble

The single-target post-processing folders load the ten selected manifests. Designer groups models by target and constructs the progressive stacking ensemble, or uses matching stored coefficients when supplied. Check the coefficient source before interpreting the output.

Output layers

BCCYield_designeroutput.txt contains model/ensemble information and design progress. BCCYield_anaoutput.txt starts with material descriptors and ends with model-performance summaries. __R__.BCCYield_anaoutput.txt is the prediction-oriented table. Other files describe coefficients, equations, feature interactions and scans. A long analysis file may therefore contain more than one logical table.

The prediction table's row ordering follows the material data used in that run. It does not include an independent, trustworthy chemical identity simply because its predicted values look plausible. Reattach the Name and parsed composition through verified row correspondence or an explicitly constructed identifier table. Do not align a shuffled prediction table to an unshuffled material list.

Actual ensemble results

Final whole-data ensemble	Number of labeled rows	R2	RMSE	MAE	Invalid count
bag0_Yield	269	0.862234	184.433034	144.752732	0
bag0_Fracture	153	0.692446	8.845106	7.130397	0

These are final **whole-data** metrics in the archived Designer tables. They include data used in individual-model construction and ensemble fitting. They must not be reported as outer-test accuracies. They are also not the oxygen-ion-conductor benchmark numbers from the preprint.

The summary headers include R2, PearsonR and PearsonR2. R2 based on residual sum of squares is not generally identical to squared correlation for arbitrary predictions. Use the metric actually named in the table, especially when comparing a raw single model, a recalibrated one-member stack and a multi-member ensemble.

Why the consensus is not a simple average

For Yield, the final intercept is about -25.3696283. Several stacking weights are negative, including approximately -0.2667594, -0.0986623 and -0.0927490; the remaining weights are positive. The archive is using a fitted linear combination with an intercept, not a probability-weighted vote.

Progressive accumulation indices start at zero, so accumulation level 0 means one member and level 9 means ten. A one-member stack can already differ from the corresponding raw single model because its intercept/slope are refitted. Increasing the number of members should be judged from the full accumulation table, not inferred from the last member alone. [E, single-target Designer stacking sections and analysis summaries.]

For new candidates, freeze this ensemble before evaluation. Re-fitting a stack on candidate placeholders is not deployment of the trained model.

2.8 Interaction summaries and feature-scan tables

.zc.txt: occurrence and coefficient-based statistics

The file bag0_Yield.zc.txt contains columns such as TermName, InteractionLevel, Min(|Z|), Ave(|Z|), Max(|Z|), InHowManyModelsAppear, HowManyAppear and WhereToFind(z@address).

InteractionLevel is the number of base descriptors in the queried set. InHowManyModelsAppear counts models containing that set, whereas HowManyAppear counts term occurrences. A descriptor may occur more than once in a model. Addresses such as 005001.txt-1-2 identify the manifest and internal chain/term location; the numerical suffixes are code-style indices, not the one-based ordinal headings in a manuscript table. [P, SI S30-S36; GD, term-analysis routines.]

In the archived Yield ensemble, Og, AveThermalExpansion and the pair AveThermalExpansion-Og appear across all ten models. In Fracture, Og appears in all ten, while AveShearModulus and its interaction with Og have lower occurrence counts. This indicates recurring representation in these fitted equations; it does not establish an independent causal intervention on temperature or modulus.

Do not silently interpret Ave(|Z|) as nonnegative. The current aggregation uses signed stacking coefficients. The archived Yield row for AveXensity contains Ave(|Z|) = -0.3241703455, despite the absolute-value notation in the header. Source inspection also shows occurrence-dependent aggregation. Consequently the field is an implementation-specific signed statistic and requires reconciliation with the intended manuscript definition before being described as a conventional absolute importance measure. [GD 6006-6037; E, bag0_Yield.zc.txt.]

The file can contain additional chain-related sections after the initial descriptor rows. A script that simply concatenates every matching name can double-count the same logical interaction. Identify the intended section and its header first.

Feature-scan / partial-dependence-style outputs

Files such as `_bag0_Yield.AveThermalExpansion-Og.txt` tabulate a model surface while other features are fixed at their reference means. They are the implementation's feature-scan/partial-dependence-style outputs. This is not necessarily an average over the empirical joint distribution of all other features, and it is not an experimentally realizable independent variation of alloy composition. [P, SI S36-S38; GD, feature-map routines.]

Typical columns include `vAveThermalExpansion`, `vOg`, the actual `AveThermalExpansion` and `Og` coordinates, the predicted target, `ModelStdDv`, sample counts and local summary statistics. The `v...` columns are normalized mesh coordinates; use the actual feature columns for a raw-unit axis. A - in an empty local-data summary is not a measured zero.

For interpretation, inspect both the predicted surface and the nearby-data counts. A visually smooth low-error surface can extend into feature combinations unsupported by the measured dataset. Independent descriptor variation may also break the correlations induced by a real chemical composition.

`bag0_Yield.onevtest.txt` and the corresponding `Fracture` file provide additional one-variable-analysis information. They are not an extra independent held-out test dataset, despite test in the filename. Read their internal headers and connect them to the feature-analysis operation rather than treating them as another benchmark.

LaTeX and Mathematica equation exports

Designer emits `.latex.txt` and `.mathematica.txt` files for further symbolic analysis. Each file records the model components and coefficients, followed by the intercept. Use the raw coefficients with the corresponding component expressions to reconstruct the numerical model; standardized coefficients support interpretation rather than substitution into the raw-unit prediction equation.

Use `.latex.txt` for mathematical typesetting and `.mathematica.txt` for Mathematica expressions. The `L=` and `M=` descriptor labels determine their respective display names. Keep the variable definitions and original numerical transform constants with the exported model. [GD, equation-export writers.]

2.9 Three performance statements that must remain distinct

Statement	Supported interpretation	Unsupported interpretation
Regressor TEST metrics	Validation performance used inside model search	Fully independent external-test performance
Designer single-target bag metrics	Whole-data fit/evaluation of the archived ensemble	The preprint's outer-fold benchmark
Dual-table predictions	Screening of a prepared candidate list	Accuracy against 43,758 measured mechanical-property labels

The preprint separately describes five-fold outer benchmarking: 20% is excluded for testing, while the other 80% is split 8:2 for training and validation, giving 64/16/20 overall. It also explicitly describes full-data ensemble models constructed without a separate test set. Preserve that distinction rather than replacing every occurrence of the word test in the manuscript. [P, p. 9 and p. 12.]

3. Example002: social-psychology satisfaction models

3.1 What this example demonstrates

For the related social-psychology study, see doi:10.31234/osf.io/waqjn_v2 [SP].

Example002_SocialPsychology demonstrates GoodRegressor on a numeric, non-materials dataset. The response is satisfaction. The folders Jpn020 and Kor020 analyze the rows coded Nation = 1 and Nation = 2, respectively. This chapter retains these archive labels. It does not infer a questionnaire, sampling frame, scale validation, or population-wide conclusion from the filenames.

The archive contains **421 observations: 201 in Jpn and 220 in Kor**. Each observation has an identifier, 20 numeric attributes including Nation, and one target. The regression uses 19 substantive base attributes; Nation is a cohort filter, not a selected predictor. The supplied target values range from 0 to 1. The questionnaire and its scoring codebook are not included in the inspected example, so the values are described on their supplied coded scale, not translated into unprovided response categories. [E2, 001_Designer/ajsp_re_atomdat.txt; 001_Designer/ajsp_re_dat.txt; 002_Curator/input.txt.]

There are two distinct analyses to keep separate. The **full-cohort analysis** builds ten models per cohort and fits a stacking ensemble to that cohort. It supports examination of fitted equations, recurring feature interactions, and model slices. The **outer-fold analysis** builds another ten models per cohort per excluded fold, fits the stack on the development rows, and evaluates the frozen stack on the excluded rows. The second analysis is the genuine held-out test workflow; the first is not.

Directory-to-workflow map

Folder	Main contents	What to read first
001_Designer	Synthetic-record representation, attribute lookup table, preliminary descriptor report	Configuration, ajsp_re_atomdat.txt, ajsp_re_dat.txt
002_Curator	Prepared numeric table and interaction-expansion report	input.txt, configuration, ajsp2gr.txt
003_Regressor/ajsp2gr	120 search directories, 2,400 stage reports, 120 selected model-chain files	A run's output.txt, then its selected chain
004_Designer/Jpn020 and Kor020	Whole-cohort ensembles, interactions, feature scans, five training/test directory pairs	Main analysis report; each pair's fold.txt and test satisfaction.se.txt
005_MLBM	Separate machine-learning benchmark inputs, scripts, logs, predictions and SHAP outputs	metrics_summary.tsv, loader settings, then fold predictions

There is **no 001_Parser step** here. The observation records are already prepared in the representation expected by Designer. The path is therefore Designer -> Curator -> Regressor -> Designer, with a separate benchmark branch. Do not invent chemical formula parsing, atom substitution, or a materials-design objective for this example.

Inventory of the regression work

Analysis branch	Cohorts	Outer folds	Models per cohort/fold	Searches	Stage reports
Full-cohort interpretation	2	Not applicable	10	20	400
Held-out outer-fold evaluation	2	5	10	100	2,000
Total				120	2,400

Every search has 20 stage summaries and a selected-model manifest. The full-cohort branch contains 20 searches and 400 stage reports; the outer-fold branch contains 100 searches and 2,000 stage reports. [E2, 003_Regressor/ajsp2gr.]

3.2 Read the observation representation before the models

Designer normally looks up properties of constituents. This example reuses that interface by assigning each observation a synthetic constituent identifier and unit content. The Atom field in the lookup table is consequently an observation key, **not an element symbol or an atomic number**. Likewise, the Curator column WifeName contains numeric identifiers in this archive; it is not a predictor selected by GoodRegressor.

The first record in 001_Designer/ajsp_re_dat.txt, with spacing simplified for display, is:

```
1 || None || 1 1 || 0.5
```

The leading 1 identifies the record. None is the dummy structure label. The pair 1 1 means synthetic constituent 1 with content 1. The final 0.5 is the supplied satisfaction response. Designer looks up the attribute vector for key 1 in ajsp_re_atomdat.txt. With exactly one unit-weight constituent, a correctly handled mean returns that observation's attribute value; it does not summarize a sample of respondents.

The small ajsp_re_strdat.txt declares the placeholder descriptor/site WW/W and a None row. UseStrName = 0 and the attribute-based descriptor declarations mean this is not a substantive structural model. Do not interpret the repeated words Atom, Materials, Valence, Cation or Structure in the shared interface as psychological variables unless the example explicitly maps them to data.

Literal fields and their roles

Fields	Role in this example	Interpretation boundary
Atom / WifeName	Observation lookup/row identifier	Preserve correspondence; do not intentionally add the identifier to setX.
Nation	Cohort selector: 1 for Jpn, 2 for Kor	Used by SubFilterX; not among the 200 GoodRegressor predictors.
AgeWife, AgeHusb, DAgeHW	Age-labelled attributes and the supplied difference field	Keep the existing coding and difference convention.
AcaWife, AcaHusb	Academic/education-labelled coded attributes	The archive does not supply their category labels.
IncomeWife, IncomeHusb, DIncomeHW	Income-labelled coded attributes and difference	Do not assign a currency or raw income unit to the fractional codes.
MarrYr	Marriage-duration-labelled attribute	Preserve the supplied numeric values and header.
F_oneflesh, F_famprefer, F_desireforind, F_selfalloc, F_psychoindiv	Five F_* score fields	Preserve these labels; their full instruments and scoring definitions are not supplied here.
avoid, assert, yield, reconcil, integrate	Five response-style-labelled score fields	These are input scores, not instructions to the optimization engine.
satisfaction	Target/response; also the model-group name	Use the coded scale in error metrics; it is not a probability model merely because values lie in [0,1].

The L= and M= strings in the Designer configuration provide LaTeX and Mathematica display labels, such as AgeHusb mapped to an A_h display. They do not replace the literal numeric field names. The unmodified reference configuration is included in reference_inputs/example002 for matching these labels to the files.

The numeric handoff is already prepared

002_Curator/input.txt has 422 lines including its header and 22 columns: identifier, 20 attributes and target. Its 421 identifiers and 20 attribute values correspond to the lookup table in the same row order. This prepared numeric table is the starting point for the Curator stage. The later Designer configuration uses the mean-only setting 2 for the singleton observation representation.

3.3 Curator: from 19 base predictors to 200 inputs

The Curator's XSpace contains all 20 attributes, including Nation. Its InteractXSpace contains the 19 non-Nation attributes. Thus the cohort indicator remains available for filtering without becoming part of the interaction pool. NameDuplicateHandling = -1, ParseData = -, ParseGroup = 3, setY = satisfaction **, and SubFilterX = - describe the shared, unfiltered table at this stage. [E2, 002_Curator/basic_properties_GoodCurator.txt, lines 1-14.]

The resulting selected predictor catalogue has the following composition:

Component	Count	Example
Original non-cohort attributes	19	AgeHusb, F_psychosindiv, assert
Pair products	171	(AgeHusb*assert)
Admitted pair ratios	10	(AgeWife/AgeHusb)
Total supplied to setX	200	Literal headers are used as predictor names.

There are 171 unordered pairs of 19 distinct base features. Ratio construction is more restricted than multiplication. In this input, the five eligible nonzero, fixed-sign fields are AgeWife, AgeHusb, MarrYr, F_selfalloc, and F_psychosindiv. The archive contains one oriented ratio for each of their ten pairs, in input order. It does not contain both directions for every pair, nor every possible division among all 19 features. Later symbolic construction can introduce additional transformations and composite interactions. [E2, Curator report and setX in 003_Regressor/ajsp2gr/Jpn020001/basic_properties_GoodRegressor.txt; GC, interaction generation.]

ajsp2gr.txt is a **report containing a dataset block**, not just a rectangular matrix. The complete-cohort Regressor input.txt files have 421 data rows and 203 columns: WifeName, satisfaction, Nation, and the 200 predictors. The outer-fold versions add Fold1 through Fold5, giving 208 columns. Those indicators are filters, not additional members of setX.

Use the extractor provided with the manual to isolate the Curator's tabular data block when creating a new handoff; do not feed the whole log to Regressor. Check the resulting header against setX. The saved Regressor inputs are the immediate ground truth for what the archived searches received. They should be retained with the model even when the Curator can regenerate equivalent features.

3.4 Regressor configuration: full-cohort and outer-fold runs

The names Jpn020001 through Jpn020010, and their Kor counterparts, identify ten full-cohort searches. The additional suffix _Fold3, for example, identifies a search whose development data exclude outer fold 3. The 020 in the directory naming must not be substituted for the final number of model terms: the selected model may have considerably fewer than 20 terms.

Settings that control this example

Setting	Archived value	Meaning here
setY	satisfaction **	Model the supplied target with no filtering bounds.
setX	200 literal names	The expanded pool just described, excluding identifier, cohort and fold indicators.
NVariableLimit	-20 -19l ... -1l	Twenty stages; decrement active additive terms while allowing hierarchical interaction expansion.

Setting	Archived value	Meaning here
PickUpPLevel	0.05	Threshold for the implemented coefficient/model significance filters.
TrainRatio	0.8	Internal training/validation split after cohort and outer-fold filtering.
NTestRobustness	100	Evaluate the fitted predictions on 100 resampled subsets; not 100 refits.
TargetCPUTimeBeforeSwap	-100 -1 !!	A 100-second run-through time target with the faster swap branch and the encoded !! mode. No positive after-swap cap is specified.
Active transforms	67	Counted before the BetaRegression boundary; not the preprint's 109-transform study setting.
BetaRegression	none	Linear-coefficient symbolic regression, not a bounded beta-response fit.

The time record does not promise a 100-second end-to-end calculation. The punctuation is interpreted by Regressor; it is unrelated to the standalone ! that disables Designer's composition optimization. See Section 1.5 for the full record semantics. Domain restrictions of roots, logarithms and divisions still apply even though the response is bounded in the supplied data.

For the full Jpn cohort, the cohort filter is:

```
SubFilterX(IfNone, "-";{Name/Min/Max:IfNoLimit,Min/Max=}*NSubFilterX) Nation 1 1
```

The corresponding Kor filter uses Nation 2 2. The Jpn outer-fold 1 configuration additionally requires Fold1 = 0:

```
SubFilterX(IfNone, "-";{Name/Min/Max:IfNoLimit,Min/Max=}*NSubFilterX) Nation 1 1 Fold1 0 0
```

These excerpts illustrate changed records, not complete runnable configurations. Preserve the fixed record order, the full 200-name setX line, the complete transform block, and the terminating records from the supplied configuration.

Actual sample counts

The full-cohort Jpn searches use 160 internal training rows and 41 validation rows; the Kor searches use 176 and 44. For outer-fold 1, Jpn has 160 development rows and splits them into 128 training plus 32 validation rows. Kor has 176 development rows and splits them into 140 plus 36. Other Jpn folds have 161 development rows and a 128/33 split. Integer counts, rather than rounded percentages alone, explain these differences.

The selected-report prediction IDs were checked against the cohort and fold input flags. In all 100 outer-fold searches, the selected model's combined training and validation rows excluded its designated outer-test rows. This verifies the stored membership, not an independent rerun of every search or a claim that every preprocessing choice was learned within each fold.

3.5 Read the stage summary before the final equation

output.txt contains the 20-stage summary and a RECOMMEND block. The recommended entry is the stage with the largest archived average resampled R2, rather than necessarily the final stage, the smallest equation, or the highest internal validation R2. In all 120 searches, the recommended entry agrees with the maximum of the 20 summary values and with the terminal file in its selected manifest.

For Jpn020001, selected stages along the trajectory are:

Stage	Active terms	Average resampled R2	Training R2	Validation R2
1	20	0.699345	0.677828	0.806252
3	18	0.715781	0.716075	0.764033

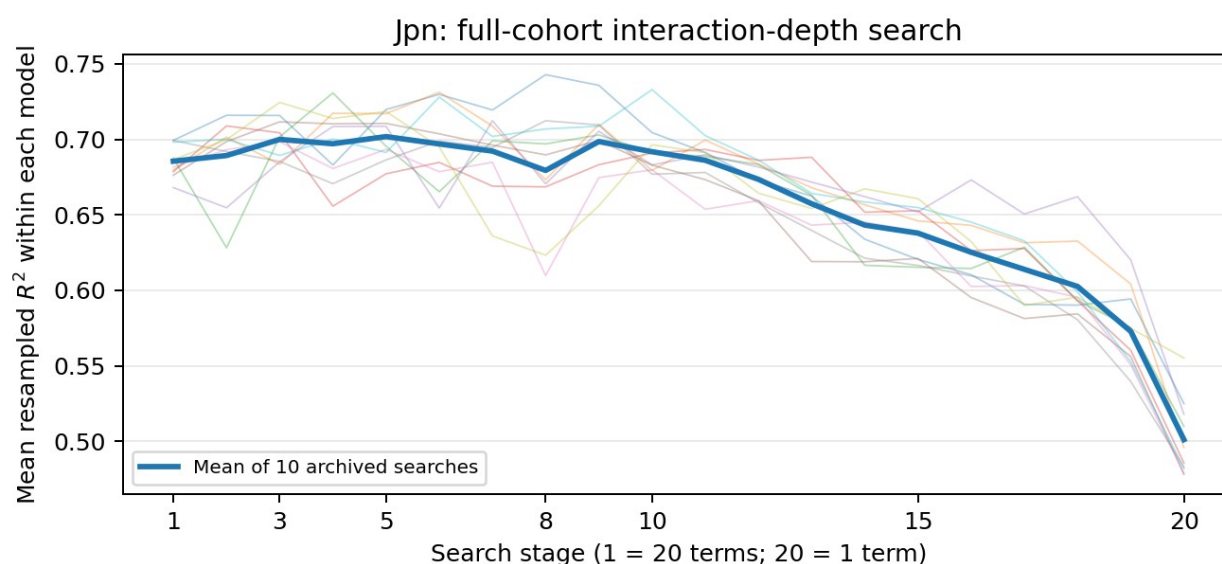
Stage	Active terms	Average resampled R2	Training R2	Validation R2
6	15	0.729790	0.710934	0.795420
7	14	0.719404	0.784609	0.500415
8	13	0.742815	0.779008	0.652128
9	12	0.735740	0.737718	0.748265
15	6	0.620513	0.640448	0.570932
20	1	0.525063	0.520547	0.545201

The chosen model is stage 8 with 13 active additive terms, saved as `output_satisfaction_8L13I.txt`. Its validation R2 is lower than the first-stage value, while its average resampled R2 is higher. That apparent contradiction disappears once the two criteria are distinguished: validation steers local search, while the reported robustness aggregate is used for this stage recommendation. The aggregate includes predictions for development rows used during fitting; it is not a replacement for the outer-fold results in Section 3.10.

The 20 selected full-cohort models

Search	Chosen stage	Terms	Average resampled R2	Validation R2
Jpn020001	8	13	0.742815	0.652128
Jpn020002	6	15	0.731188	0.669103
Jpn020003	4	17	0.730705	0.733155
Jpn020004	2	19	0.708874	0.746680
Jpn020005	7	14	0.712298	0.693070
Jpn020006	3	18	0.711572	0.833924
Jpn020007	3	18	0.698405	0.848335
Jpn020008	8	13	0.712315	0.733815
Jpn020009	3	18	0.724451	0.782643
Jpn020010	10	11	0.732896	0.714688
Kor020001	9	12	0.712223	0.763636
Kor020002	2	19	0.696470	0.731323
Kor020003	7	14	0.703180	0.563496
Kor020004	3	18	0.717993	0.582068
Kor020005	5	16	0.683172	0.647962
Kor020006	3	18	0.690716	0.624760
Kor020007	2	19	0.717689	0.727777
Kor020008	3	18	0.681834	0.716184
Kor020009	4	17	0.702164	0.789923
Kor020010	3	18	0.709091	0.750120

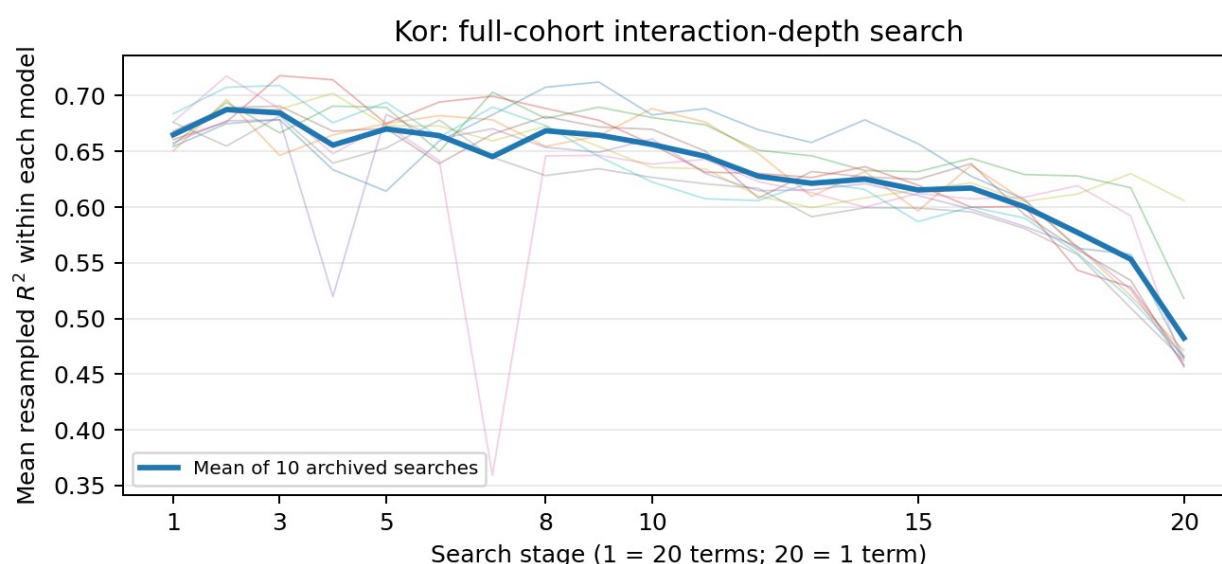
The selected Jpn models have 11-19 active terms, and the selected Kor models have 12-19. These are counts of the final additive components, not counts of distinct raw score fields, scalar operations, or interaction levels. A component can contain several transformed and nested base features.



Internal model-selection criterion; not independent outer-test performance.

Jpn full-cohort stage trajectories

Figure 3.1. The ten Jpn trajectories and their mean, calculated from the archived stage-summary records. Increasing stage number decreases the allowed term count. The vertical quantity is an internal model-selection criterion, not held-out test accuracy.



Internal model-selection criterion; not independent outer-test performance.

Kor full-cohort stage trajectories

Figure 3.2. The corresponding Kor trajectories. The curves summarize these saved searches only; they do not establish a general psychological complexity ranking between cohorts.

3.6 Worked output: Jpn020001, stage 8

Open 003_Regressor/ajsp2gr/Jpn020001/output_satisfaction_8L13l.txt. Long reports include search progress before the selected result, so locate FINAL_ANSWER rather than reading the first number that resembles a score. In this file, FINAL_ANSWER begins at line 711, the coefficient table at line 726, TEST at line 903, and ROBUSTNESS at line 952.

Training, validation and resampled summaries

Report block	Rows / repetitions	R2	RMSE	MAE
FINAL_ANSWER, training	160 observations	0.779008	0.143884	0.110888

Report block	Rows / repetitions	R2	RMSE	MAE
TEST, meaning validation	41 observations	0.652128	0.178455	0.144479
ROBUSTNESS, Ave	100 subsets of 41 rows	0.742815	0.150454	0.116277

All errors are on the supplied satisfaction scale. The model has 13 active terms, an intercept and 146 residual degrees of freedom: 160 -13 -1. Its report prints full_Frequentist = 1; this means that the implementation's significance conditions were satisfied. It does not establish causality, external validity, or absence of collinearity.

The numerical form is an intercept plus 13 coefficient-weighted components. The first printed component is $(\text{erf-2}(\text{AgeHusb} * \text{assert}))^1$, whose transform label should be looked up in the original configuration. Here erf-2 means the erf row with $b = 0.01$, not an error function raised to the power minus 2. With $a = 0$ and $c = 1$, this component is $\text{erf}(0.01 * \text{AgeHusb} * \text{assert})$.

First component field	Printed value, rounded	How to use it
Fitted coefficient	6.238546	Multiply the component value in the actual model.
Standardized coefficient	2.407953	A scale-adjusted contribution used in model reporting; not a probability or direct intervention effect.
Standard error	1.026523	The report's coefficient uncertainty estimate under its fit.
t value	6.077359	Statistic used by the implemented significance check.
p value	1.10596e-8	Printed numerical significance value, not a multiple-search causal guarantee.
Lower / upper coefficient bounds	4.108264 / 8.368827	The report's interval fields.
VIF	103.715251	The same output records substantial dependence among model components despite the significance flag.

An individual coefficient cannot be read in isolation as the marginal relationship between one raw score and satisfaction: AgeHusb and assert can appear in other components as well. The full chain, all coefficients and the intercept are required for numerical evaluation.

Do not confuse the two dispersion quantities

In this report, robustness R2 has Min = 0.580058, Ave = 0.742815 and Max = 0.880566. The corresponding standard deviation is approximately 0.061903. It measures variability of the metric across the resampled subsets.

By contrast, Designer's per-observation ModelStdDv or bag0_satisfaction_stdev describes dispersion among the individual-model predictions for one observation. It is a different quantity from the standard deviation of robustness R2. Neither quantity is automatically a calibrated prediction interval.

The report's subsequent INTERACTION section records the expanded representation used to reconstruct later hierarchical components. The selected report by itself is not necessarily a standalone predictor: retain its predecessor reports and its training-time transform configuration.

3.7 Selected manifests and portable reconstruction

A selected file such as 003_Regressor/ajsp2gr/Jpn020001.txt is a small manifest, not the equation text. Its first records name the target, identify beta mode as none, and point to basic_properties_GoodRegressor.txt. The chain_filenames block then lists stage 1 through stage 8 for this particular selected model. A different search can stop at a different stage.

The supplied manifests contain absolute paths beginning with the original /work/... directory. Moving the archive to another computer does not make those paths valid automatically. In a working copy, replace the old root consistently in the configuration reference and every chain path; preserve the rest of the path and the stage order. Do not point a _Fold1 manifest to the similarly named full-cohort directory merely because its files are easier to find.

CodeOnly/Utility/find_maxR2.py reads the recommendation in each run's summary and generates the selected-model file for Designer. Run it in a working copy with the intended search directories; the example manifests show the saved selections.

Designer also produces LaTeX and Mathematica exports of the selected models. Section 2.8 explains their component, coefficient and intercept fields.

3.8 Designer: full-cohort ensemble analysis

The main Jpn and Kor post-processing configurations load their respective ten selected manifests. Their RegModelFilename line ends with a standalone !. In this context, ! selects collection/analysis rather than composition optimization. Preserve it: replacing synthetic observation identifiers with other identifiers would not constitute a valid intervention on a person.

The important input combination is:

```
MaterialsFilename(...)/HeadExists(...)/PostTagNames(...) ajsp2gr_dat.txt 0 satisfaction
RegModelFilename(...) Jpn020001.txt ... Jpn020010.txt !
AnalyzeInteractionLevel 2 10
SkipMinMaxKurtoInAnalyzeInteraction(Yes1No0) 2
```

The ellipsis above abbreviates unchanged labels and the member list for discussion only. Use the full supplied reference input to run the program. AnalyzeInteractionLevel and the mean-only setting are distinct controls: the first requests interaction analysis, while the second ensures the observation-attribute representation is interpreted using the mean-only branch.

Other shared-interface records remain present, including TargetVReg = 1, TargetStdDvReg = 0.5, ZDistWeight = 1, and UseOffset = 1. Their presence does not mean this archive has optimized a respondent to a desired satisfaction value. The saved output used here is the direct model/ensemble analysis. In particular, TargetStdDvReg is not the optional outer-test TestModelStDevLimit record described below.

Prediction table: one row per observation

The full-cohort __R__.ajsp_re_anaoutput.txt has 201 Jpn or 220 Kor data rows and 13 columns:

Column group	Number	Meaning
reg0_satisfaction through reg9_satisfaction	10	The individual selected-model predictions in the manifest-list order.
bag0_satisfaction	1	The final fitted stacking-ensemble prediction.
bag0_satisfaction_stdev	1	Model dispersion for that observation.
satisfaction	1	Supplied observed target for comparison.

The prediction table does not carry an independent identifier column. Reattach IDs from the matching ajsp2gr_dat.txt selection in the same verified order. For a fold, apply the fold's row-selection rule before attaching IDs; joining the first 41 original rows to fold 1's 41 predictions would be wrong.

The detailed analysis report also contains a ModelName metric table and a final StackingEnsemble coefficient block. These are different logical tables within the same text file. R2, PearsonR and PearsonR2 are separate fields. Use residual-based R2 when quoting the reported R2; do not substitute squared correlation.

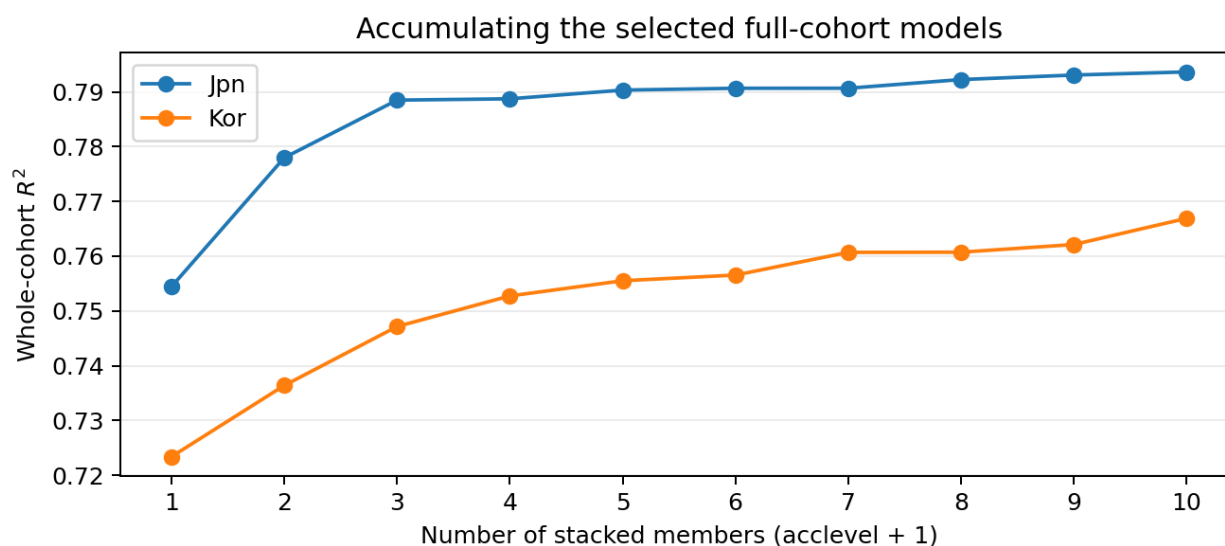
Fitting the ensemble is not taking a simple average

The stack has an intercept and ten coefficients. The Jpn final intercept is approximately -0.0489841; its member weights include positive and negative values, for example 0.3574244 for member 0 and -0.0582911 for member 3. Thus bag0_satisfaction is not the unweighted mean of reg0 through reg9. Member order and the intercept matter. [E2, 004_Designer/Jpn020/ajsp_re_anaoutput.txt, final 11-line block.]

acclevel_0 refers to fitting the ensemble from its first member, while acclevel_9 refers to all ten. This index is **not** Regressor's interaction-depth stage. Even a one-member stack may include a fitted intercept and calibration coefficient, so it should not automatically be treated as numerically identical to the raw member prediction.

Full-cohort ensemble	Labeled rows	R2	RMSE	MAE	Invalid
Jpn	201	0.793644	0.138724	0.106879	0
Kor	220	0.766905	0.123113	0.096828	0

These are whole-cohort fitting/evaluation metrics. All observations in the respective cohort participate in this branch's construction or stack fitting, and invalid count is 0 in both final full-cohort summaries. They must not be captioned as independently tested predictive performance.



Ensemble fitting/evaluation on the same cohort; not held-out test accuracy.

Full-cohort ensemble accumulation

Figure 3.3. Whole-cohort R2 as members are accumulated. The figure describes ensemble fitting, not a learning curve on an untouched test set. The actual held-out fold results are reported separately with their retained coverage.

3.9 Interaction tables and model-slice outputs

Read .zc.txt as an equation-structure summary

bag0_satisfaction.zc.txt summarizes recurring base-feature sets across the selected models. InteractionLevel = 1 denotes one base feature; level 2 denotes a pair co-occurring within a model component. InHowManyModelsAppear counts models containing that feature set, whereas HowManyAppear counts occurrences across components. These are not the same count.

The initial table contains the following illustrative entries, rounded from the files:

Cohort	Feature set	Level	Models	Occurrences	Printed Ave(abs Z)
Jpn	AgeHusb	1	10	40	0.720953
Jpn	assert	1	10	51	0.540328

Cohort	Feature set	Level	Models	Occurrences	Printed Ave(abs Z)
Jpn	yield	1	10	47	0.444657
Jpn	MarrYr-assert	2	10	23	0.103017
Jpn	IncomeHusb-yield	2	9	31	0.508124
Kor	avoid	1	10	49	0.552505
Kor	AgeHusb	1	10	37	0.435802
Kor	F_psychoidiv-assert	2	10	16	0.075733

For example, Jpn AgeHusb appears in all ten models, with 40 occurrences across terms. That means repeated representation in this fitted symbolic ensemble; it does not by itself establish that changing age causes a particular satisfaction change. A frequent pair can have a smaller aggregated coefficient statistic than a less frequent pair. The source's chain-selection logic and its coefficient aggregation should not be replaced by an improvised sorting rule.

The headings $\text{Min}(|Z|)$, $\text{Ave}(|Z|)$ and $\text{Max}(|Z|)$ refer to the implementation's coefficient statistics. As discussed in Section 2.8, signed stacking weights enter the aggregation; do not assume the $\text{Ave}(|Z|)$ label is a conventional nonnegative feature-importance definition. Additional chain sections later in the file repeat some entries. Use the initial table or a specifically identified chain section rather than adding every repeated row together. [GD, lines 6006-6037 and interaction-chain reporting.]

One-variable and two-variable slices

A file such as `_bag0_satisfaction.F_oneflesh.txt` contains 101 mesh points, plus its header. `_bag0_satisfaction.F_oneflesh-reconcil.txt` in Jpn contains 10,201 mesh rows, corresponding to a 101-by-101 grid. Those counts are **not additional respondents or test observations**. The separate `bag0_satisfaction.onevtest.txt` also contains analysis sections; test in that filename does not mean an independent dataset.

Field	What is plotted or summarized
vF_oneflesh, or another v... field	Normalized mesh coordinate.
F_oneflesh, or the corresponding bare field	The coordinate in the supplied numeric coding. Use this for an unnormalized axis.
satisfaction	Ensemble prediction at the displayed coordinates, with other features at reference values.
ModelStdDv	Dispersion across the member models at the mesh point.
Nsample, Min, Ave, +-, Max	Local observed-data counts and summary values associated with the mesh location.
Nsample(R), Min(R), Ave(R), +-(R), Max(R)	Corresponding summaries of regressed responses.
MeanAE, MaxAE	Local absolute-error summaries produced by the routine.

For the first Jpn F_oneflesh mesh row, the coordinate is 0, the surface prediction is approximately 0.0768711, ModelStdDv is 0.0956302, and Nsample = 2. The local observed mean is 0.25; the local mean regressed value is approximately 0.152199. These differ because a model slice with other features fixed is not the same operation as averaging the actual observations or averaging their individually evaluated predictions. The next grid point has - in the local-data fields: that indicates no reported local summary, not a zero measurement. [E2, 004_Designer/Jpn020/_bag0_satisfaction.F_oneflesh.txt, lines 1-3; GD, feature-map writer near 6321.]

These are **fixed-reference feature scans**, not necessarily population-averaged partial dependence. Varying AgeHusb while fixing AgeWife and DAgeHW, or varying a fractional category code continuously, can produce combinations outside the represented observations. Inspect local support, the supplied scale coding and numerical validity before interpreting a curve. The archive does not establish that these slices are feasible interventions or causal effects.

3.10 Genuine outer tests: select folds, freeze the stack, report coverage

This example provides a second layer of evaluation beyond the legacy TEST block. Its outer-test rows are excluded from the corresponding Regressor searches and from stack fitting. The `_Fold1` searches feed `Fold1_train` and `Fold1_test`; the full-cohort searches do not substitute for them.

Decode fold.txt

The Jpn fold 1 development directory contains:

```
NFolds TakeFold Train0Test1 Straight0Rotate1
5 1 0 1
```

The corresponding test directory contains:

```
NFolds TakeFold Train0Test1 Straight0Rotate1
5 1 1 1
TestModelStDevLimit= 0.5
```

`NFolds = 5` and `TakeFold = 1` select the first outer fold. `Train0Test1 = 0` keeps its complement; 1 keeps the fold itself. `Straight0Rotate1 = 1` uses rotating, row-order assignment. With zero-based row position i within the cohort, the fold is $(i \% 5) + 1$. This is not random shuffling and is not computed from the numeric identifier: IDs can have gaps. [GD, fold reader near 9378-9403 and row-selection code 6641-6656.]

Jpn test-fold sizes are 41,40,40,40,40; Kor has 44 in each fold. The development sets have 160 or 161 Jpn rows and 176 Kor rows. Within each development set, ten selected symbolic models are constructed with their own internal training/validation operation. Designer then fits the stack using those development rows.

How the archived script keeps test coefficients fixed

`run_traintest.sh` prepares per-fold training directories using the fold-specific manifests. It then builds test directories from the training setup and copies the last 11 lines of the training analysis report to `satisfaction.se.txt`: one intercept and ten weights. The test evaluation loads these coefficients rather than fitting another stack to its held-out observations. [E2, 004_Designer/Jpn020/run_traintest.sh, lines 456-467, and Kor counterpart.]

The test and training directories contain canonical names such as `Jpn020001.txt`. Their **contents**, not just the copied basename, identify `_Fold1` model dependencies. The audit checked that all ten test coefficient files match the final block in their respective training reports and that the selected Regressor memberships exclude those test IDs. Keep the member order unchanged when copying a coefficient file.

This supports the intended stored separation of model fitting and outer evaluation. It does not prove that every data-preparation choice was fitted inside the folds: a common curated feature representation exists before fold filtering. Record that representation and its provenance rather than claiming an uninspected fully nested preprocessing procedure.

The dispersion screen changes the denominator

The test-side `TestModelStDevLimit = 0.5` is active. The evaluation branch inserts a nonfinite ensemble result when model dispersion exceeds this limit; genuinely nonfinite numerical predictions can also be invalid. The metric table excludes invalid predictions. The result is **conditional test performance on retained rows**, not necessarily performance on all rows assigned to that fold. [GD, lines 7126-7142.]

Cohort / fold	Assigned	Retained	Rejected	R2	RMSE	MAE
Jpn / 1	41	40	1	0.562357	0.204776	0.173363
Jpn / 2	40	40	0	0.386462	0.230755	0.187063
Jpn / 3	40	38	2	0.638999	0.161933	0.123334
Jpn / 4	40	40	0	0.311118	0.266086	0.211849
Jpn / 5	40	36	4	0.307738	0.238199	0.194688

Cohort / fold	Assigned	Retained	Rejected	R2	RMSE	MAE
Kor / 1	44	43	1	0.412812	0.193535	0.152152
Kor / 2	44	42	2	0.183012	0.220406	0.173706
Kor / 3	44	38	6	0.619740	0.144766	0.113808
Kor / 4	44	44	0	0.262663	0.228516	0.167048
Kor / 5	44	43	1	0.213300	0.208776	0.170210

The metrics in this table were independently recalculated from the stored prediction rows and agree with the archived summaries. The displayed R2 is the R2 field, not PearsonR2. Rejected here means excluded from the final metric calculation, encompassing the archive's dispersion and numerical-invalid outcomes; it is not a manually removed respondent from the original dataset.

Cohort	Total retained	Coverage	Mean fold R2	Mean fold RMSE	Mean fold MAE
Jpn	194/201	96.52%	0.441335	0.220350	0.178059
Kor	210/220	95.45%	0.338305	0.199200	0.155385

The mean row averages the five fold-specific metrics with equal fold weight; it is not R2 recomputed after pooling all predictions. Coverage is computed from total retained and assigned observations. Report both. A statement such as "Jpn outer-test R2 is 0.4413" is incomplete without the 0.5 dispersion rule and 194/201 retained coverage.

Do not recalibrate satisfaction.se.txt on test responses, tune the dispersion cutoff on the same test outcomes without declaring that selection, or replace invalid predictions with zero. The companion evidence retains the invalid counts and observation IDs. Diagnostic reconstruction from the saved member columns also shows that rejected predictions are not harmless blanks; finite rejected values can differ greatly from the observed scale, and some member outputs are nonfinite. A full-coverage comparison requires an explicitly specified failure-handling policy, not silently dropping difficult rows.

Safe rerun boundary

The archived helper script contains machine-specific source paths and expects a usable GoodDesigner.x. It also has RECREATE_TRAIN_DIRS = 1 and RECREATE_TEST_DIRS = 1, enabling replacement of existing fold directories. These are not read-only defaults. Do not run that script in the original archive to "check" the results. Prepare a separate working copy, inspect the path and directory-recreation controls, compile the appropriate executable, validate manifests and coefficient order, and only then launch the intended operation.

3.11 Read the benchmark branch as it was actually run

005_MLBM is a separate implementation, with one satisfaction.xlsx workbook for each cohort. The Jpn workbook has 201 observations and the Kor workbook 220; both have 22 columns. Their identifiers and target order match the corresponding cohort tables. Eight benchmark models have saved five-fold prediction files: Ridge, ElasticNet, RandomForest, XGBoost, LightGBM, MLP, PhySO and EQL. Do not add a PySR result merely because a script or environment contains optional Julia/PySR-related code.

Inputs are not identical to GoodRegressor's inputs

The benchmark's ID_LIKE_COLS is ['CompositionName']. That name is absent from these workbooks, whose identifier is **WifeName**. The loader therefore retains WifeName, Nation, and the 19 substantive attributes, giving **21 numeric predictors**. This is confirmed by both the source loader and the saved SHAP metadata's feature list. GoodRegressor, in contrast, selects 200 predictors formed from the 19 substantive attributes and excludes WifeName and Nation. [BM, benchmark_v14.py, line 112 and lines 1399-1465; E2, 005_MLBM/ajsp2gr_Jpn/out/shap/satisfaction_shap_RandomForest_metadata.json, and Kor counterpart.]

Within each cohort Nation is constant, but WifeName is a varying row identifier. Its inclusion is an actual input difference that should be reviewed; this manual has not silently removed the column or recalculated the benchmark. Combined with GoodRegressor's conditional prediction coverage, it prevents treating the archived numbers as a fully matched, identical-input comparison without further qualification.

Outer folds and inner selection

The benchmark script uses its RoundRobinKFold for the five outer folds. For estimators with an external hyperparameter space, it uses four-fold round-robin RandomizedSearchCV within the development data, with R2 scoring. Models whose specification has param_space = None use their own internal-search fit instead of that same external randomized-search loop. Numeric preprocessing is part of the model pipeline. [BM, RoundRobinKFold nearline 50; nested_cv_oof fromline 1218; build_preprocessor.]

The launcher schedules eight models times five folds and sets a configurable n500 hyperparameter regime. The label n500 is not the actual number of respondents: the workbooks contain 201 and 220 rows. Cluster partition names, module loads, Python setup and resource counts in the archived shell script describe its original execution environment, not universal hardware requirements for GoodRegressor.

Archived five-fold summaries

Model	Jpn R2	Jpn RMSE	Jpn MAE	Kor R2	Kor RMSE	Kor MAE
Ridge	0.34846	0.24156	0.17932	0.39706	0.19376	0.13888
ElasticNet	0.33686	0.24296	0.18712	0.40336	0.19288	0.13624
RandomForest	0.26590	0.25224	0.18136	0.38030	0.19602	0.13926
XGBoost	0.06002	0.29014	0.21826	0.20562	0.22392	0.16504
LightGBM	-0.14168	0.32090	0.23628	0.15252	0.22984	0.16410
MLP	-0.13638	0.31744	0.23834	0.30368	0.20940	0.15860
PhySO	-0.01892	0.30424	0.25872	0.02296	0.25018	0.19844
EQL	-0.96052	0.41092	0.30324	-0.91168	0.34822	0.25166

These values reproduce metrics_summary.tsv, rounded here to five decimals. The companion postprocessor takes the final matching metric line from each fold log, then computes means and sample standard deviations; the log metrics are printed to limited precision. Recalculating from the saved CSV predictions can therefore differ slightly in the last decimal. Each model has five saved fold results, and those prediction files cover the complete assigned cohort rows in this archive.

File dictionary for benchmark analysis

File or directory	What it contains	Practical check
satisfaction.xlsx	Numeric input data for this benchmark	Inspect the actual header against ID_LIKE_COLS before running.
model_logs/*_fold*.log.txt	Per-model/per-fold execution and metric logs	Keep one intended completed log per model/fold when aggregating.
out/ satisfaction_predictions_*_fold*.csv	Two-column true/predicted responses for one outer fold	Restore IDs with that fold's row-order rule, not a sequential whole-cohort join.
metrics_summary.tsv	Means, sample standard deviations and fold counts	Check n_folds_found = 5; duplicated logs can be double-counted by the supplied merger.
predictions_wide.tsv	Per-model true/predicted columns after concatenating fold files	Its row order is fold concatenation, not necessarily the original workbook order.
out/best_params	Hyperparameter records from the fold fits	Retain them with the model/version and fold definition.

File or directory	What it contains	Practical check
out/shap	Final-refit explanations and their metadata	These explain full-data refitted models, not a new independent test.

SHAP outputs: read the metadata, not only the picture

The archive contains SHAP CSVs, metadata and summary images for RandomForest, XGBoost and LightGBM in each cohort. The `--final-refit-only` mode reads saved fold hyperparameter records and fits a model to the full cohort; this step does not rerun the outer cross-validation. SHAP therefore belongs to a full-data explanation branch, separate from the held-out predictions.

The saved metadata explicitly says that SHAP values explain the **inner tree-regressor output after the target transformation**, not the inverse-transformed satisfaction scale. Its `sample_index` is a zero-based row position after loader filtering, not `WifeName`. Preserve that scale and identifier distinction when comparing a SHAP plot to a symbolic-model slice. The feature list also visibly contains `WifeName`; any interpretation of this benchmark should acknowledge that input before attributing the entire plotted importance to substantive score fields.

Appendix A. Output-file dictionary

The table covers the recurring outputs present in Example001 and the principal temporary files encountered in the source. A prefix such as BCCYield or Yield changes with the input/task.

File or section	How to use it
Parser *_log.txt	Parsing diagnostics; reconcile failures and successful material counts.
Parser *_output.txt	Intermediate parsed formula/block representation and aggregate diagnostics.
Parser *_dat.txt	Parsed-material handoff plus appended target/postscript values.
Designer *_anaoutput.txt	Descriptor/material table, and with loaded models, additional performance summaries. Identify each logical table.
Designer __R__.*_anaoutput.txt	Model/ensemble predictions and dispersion fields aligned to material row order.
Designer *_designeroutput.txt	Main model/stacking report and composition-search progress when enabled.
Curator report, e.g. BCCYield.txt	Preamble, rectangular DATASET, then summaries; not a pure TSV throughout.
Regressor input.txt	Prepared numeric table with the literal curated header.
Regressor output_<target>_<stage>.txt	Full search-stage record, final fit, training/validation rows, robustness and interactions.
Regressor FINAL_ANSWER	Locate the final chosen fit within one stage, not necessarily the selected depth for the ensemble.
Regressor TEST	Legacy name for validation data/results.
Regressor ROBUSTNESS	Fixed-model resampling diagnostic, with minimum, mean, maximum and standard deviation of the metrics.
Small 005001.txt-style manifest	Target/configuration and preceding-stage dependencies for one selected model.
Target .se.txt	Saved progressive stacking coefficients in the loaded model order.
Target/index .latex.txt	LaTeX representation of the symbolic components, coefficients and intercept.
Target/index .mathematica.txt	Mathematica representation of the symbolic components, coefficients and intercept.
bag*.zc.txt	Descriptor/interacting-set occurrences and implementation-specific coefficient statistics.
bag*.onevtest.txt	One-variable/model-analysis information; not an independent test set.
_bag*.<feature-or-pair>.txt	Feature-scan mesh, predictions, dispersion and nearby-data summaries.
<species>.nb.txt	Optional explicit replacement-neighbor input; may alter apparent configuration behavior.
fold.txt	Optional row/fold analysis selection; not a complete nested-validation pipeline.
*.temporary.txt, *.reg, *.pr.txt, smt, worker files	Intermediate coordination/working artifacts; isolate job directories and retain final dependencies before cleanup.

A.1 Additional files in Example002

File / pattern	Purpose	Distinction to retain
ajsp_re_atomdat.txt, ajsp2gr_atomdat.txt	Observation attribute lookup	Synthetic observation keys, not chemical elements.
ajsp_re_dat.txt, ajsp2gr_dat.txt	Singleton record, unit content and target	Preserve the cohort-specific row order.
Jpn020001.txt, Kor020001.txt	Selected whole-cohort model-chain manifests	Require original configuration and all predecessor stages.
*_Foldk.txt	Model selected while excluding outer fold k	Verify file contents after renaming to canonical member names.
fold.txt	Designer development/test membership and optional dispersion limit	Rotating assignment uses cohort row position.
satisfaction.se.txt in Foldk_test	Frozen development-fitted stack	One intercept plus ten ordered weights; do not refit on test responses.
__R__.ajsp_re_anaoutput.txt	Whole-cohort model predictions	Full-cohort fit, not outer-test prediction.
Foldk_test/ __R__.ajsp2gr_anaoutput.txt	Held-out predictions with validity/dispersion behavior	Report assigned and retained counts.
metrics_summary.tsv	Separate benchmark fold-mean metrics	Its predictors and coverage differ from the archived GoodRegressor setup.
out/shap/*_metadata.json	Explanation scale, feature list and row indexing	Full-refit, transformed-target explanations; not outer-test metrics.

Appendix B. Source map and documentation policy

B.1 Where the implementation is defined

Topic	Primary source location
Parser parameter order, atom blocks	GP 217-265
Curator parameter order	GC 103-234
Composition-based selection	GC pick_data, approximately 249-385
Designer configuration and descriptor declarations	GD 3430-4050; descriptor continuation near 4107
Atomic-statistic weighting and zero handling	GD get_mamssk, 2019-2093
Ensemble prediction and model dispersion	GD calc_bag_chain, 3209-3233
Model-chain reading	GD read_reg_chain, from 4944
Interaction aggregation	GD 6006-6037
Group-indexed target access	GD material-processing loop near 7783
Equation-string formatting	GD chk_eq, get_powerstring, get_powerstringM and equation-export writers
Regressor parameter order / transform boundary	GR 1290-1563
Scalar numerical functions	GR get_v, 900-940
Fixed-model robustness and standard-deviation output	GR, robustness routine and final square-root conversion

Additional implementation references for Example002

Topic	Primary source location
Mean-only mode and zero retention	GD 2019-2093 and 9455-9470
Rotating fold membership	GD 6641-6656; fold reader near 9378-9403
Test-side dispersion rejection	GD 7126-7142
Feature-scan output headings	GD near 6321; nearby observed/regressed-column lookup
Transfer of frozen stack to test directory	E2, 004_Designer/Jpn020/run_traintest.sh, 456-467
Benchmark identifier exclusion list	BM line 112; numeric loader1399-1465
Outer and inner benchmark split logic	BM RoundRobinKFold; nested_cv_oof from 1218
SHAP scale and actual feature list	E2, both benchmark out/shap/*_metadata.json files

B.2 Where the preprint supplies the conceptual explanation

The main-text Methods on pp. 24-27 explain lexicographic sampling, swap, transit, pick and repeated model construction. SI S3-S9 explains the five-step workflow and ensemble post-processing. SI S10 lists the study's scalar transforms; it is not a substitute for counting the active transforms in an actual configuration. SI S30-S38 explains interaction sets/chains and feature scans. SI S41-S43 explains offset-anchored materials prediction and standardized target distance.

The outer train/validation/test protocol is described on main-text p. 9; the full-data final-ensemble construction without a separate test set is described on p. 12. The manuscript and its illustrative older

figures contain legacy test usage in some individual-model contexts. This manual follows the author's clarification for program outputs while preserving the genuinely excluded outer-test meaning.

B.3 Source-versus-documentation distinctions

Numerical examples refer to the supplied saved calculations. Input descriptions incorporate the revised settings and definitions documented here. When adapting an example, retain its input tables, selected-model files and stacking coefficients alongside the numerical results.

Use the filenames and function names in this source map to locate the relevant operations. Line numbers identify the reference source and can move when code is edited. Keep the code version, active transformation catalogue and configuration with any newly generated results.

Appendix C. Calculation-speed features

GoodRegressor combines three acceleration mechanisms that address different sources of cost. **Cholesky updates** reduce repeated linear-algebra work when nearby candidate models are fitted. **On-the-fly learning** limits the feature/transform trials explored in a refinement pass according to the evolving fit. **Active MPI** reallocates completed ranks to ranks with unfinished work. These mechanisms complement the time-limited, lexicographically ordered initial search rather than replacing its model-selection objective.

The starting presets in Section 1.5, `-100 -1 !!` and `-1000 -1 !!`, enable the faster swap branch and on-the-fly mode. Active MPI is part of the parallel coordination workflow; `!!` is not a separate switch that turns MPI on.

C.1 Cholesky decomposition and incremental updates

For a candidate model whose symbolic components form the columns of a design matrix X , linear-coefficient fitting solves a least-squares problem. The implementation standardizes the relevant quantities and factorizes the Gram matrix:

```
G = transpose(X) * X = L * transpose(L)
G * beta = transpose(X) * y
```

The Cholesky factor L permits the coefficient solution through triangular systems. During a swap or scalar-transform trial, much of the candidate design matrix is unchanged. Rebuilding and refactorizing the entire Gram matrix for every such trial repeats work unnecessarily.

The implementation therefore caches the standardized design matrix and its Cholesky factor. If column j changes from x_j to $x_j + d$, the new Gram matrix can be written as:

```
u = transpose(X) * d
s = transpose(d) * d
G_new = G + u * transpose(e_j) + e_j * transpose(u)
        + s * e_j * transpose(e_j)
```

Here e_j selects the changed column. The code represents this correction using a two-dimensional symmetric matrix, decomposes that small matrix, and applies rank updates or downdates to the existing factor. With several changed columns, it can apply successive column updates. When too many columns change, or cached information is unsuitable, it calculates a fresh factorization instead. [GR, `cholesky_update_one_column`, `cholesky_update_multiple_columns`, `fit_any`, approximately 2595-2869.]

This reuse accelerates the repeated fitting of closely related candidate equations. The faster-swap option is selected by a **negative first value** in `TargetCPUTimeBeforeSwap`; the code uses the magnitude as the run-through time target. The numerical fitting path also contains fresh-factorization checks. Cached updates do not remove the need for a numerically usable design matrix or make singular, redundant components harmless.

The practical benefit depends on the dataset, term count and how much of the candidate model changes. This mechanism reduces repeated work; it does not imply a fixed speed-up factor or an exhaustive search of all possible models.

C.2 On-the-fly learning: adapt the explored feature and transform scope

The `!!` preset sets `useSomeShotMode = true`. In this mode, a refinement pass does not indiscriminately explore every feature position and every associated scalar-transformation trial. Instead, the scope of trials is adjusted while the model is being trained.

Descriptor swaps and scalar transformations use significance-ordered component lists. Descriptor replacement starts from the less significant side, while scalar-function switching starts from the more significant side. The current coefficient statistics establish a working boundary in that ordered list. As improved candidates are accepted and the coefficient statistics change, the boundary is updated for subsequent trials. The source records these bounds through variables such as `NNend` and log entries

containing NNendUpdated. [GR, swap_descriptors and switch_funcdtype branches using useSomeShotMode.]

This is **on-the-fly learning of the exploration scope**, not the addition of a separate neural network. The evolving model helps determine where to spend the next refinement effort. Features and function types can be considered in a restricted set of combinations in one pass rather than sweeping the complete feature-by-transformation space regardless of the current fit.

The original Functypes catalogue remains the definition of the available transformations. A shortened pass does not redefine that catalogue or permanently remove unvisited functions from the saved input. It changes which trials are attempted during refinement. This distinction matters when comparing runs with the same catalogue but different speed settings: their visited candidate sets, and hence selected equations, can differ.

Useful log phrases include:

```
swap_descriptors: useSomeShotMode
switch_funcdtype: useSomeShotMode
NNend
NNendUpdated
```

For ordinary use, the two presets in Section 1.5 are the simplest entry point. The 100-second version gives a shorter initial search budget; the 1000-second version allocates more time to the initial ordered search. Both retain the on-the-fly refinement mode.

C.3 Active MPI: LUHMANN followed by HEGEL

The parallel workflow has two complementary phases. The names **LUHMANN-AUTOPOIESES** and **HEGEL** are literal labels in the implementation and logs.

LUHMANN-autopoieses: independent regional work. The lexicographically ordered model space is distributed among MPI ranks. Each rank explores its assigned region and develops its local candidate through search/refinement. During the independent refinement phase, the ranks work on their own models. Ranks need not finish at the same time because their candidate models and remaining trials can differ.

HEGEL-MasterSlave: completed ranks help unfinished ranks. At coordination points, the program gathers completion status. When some ranks have completed their own work while others have not, completed ranks do not simply remain idle. The unfinished ranks become masters of cooperative groups, and completed ranks are assigned to those groups as helpers, called slaves in the code and its logs. They take portions of the remaining descriptor-swap or function-switch trials, working with the appropriate master's candidate state.

The coordinating rank distributes available helpers among the unfinished groups. Within each group, candidate evaluations are shared and the search continues for the master's model. Group membership is reconsidered as ranks finish. The coordination cycle terminates when all ranks have completed the relevant work. [GR, completion-state gathering, Hegel grouping and LUHMANN/HEGEL dispatch, approximately 7780-8020.]

```
Initial lexicographic distribution
-> each rank explores and refines its regional candidate
-> LUHMANN: independent local work
-> gather completion status
-> HEGEL: completed ranks assist unfinished masters
-> regroup as work completes
-> finish when all ranks are done
```

This active redistribution reduces the idle time that would otherwise arise when a fixed MPI partition leaves some ranks waiting for slower ranks. It concerns the remaining search/refinement tasks; it need not mean that the program moves an entire original lexicographic interval between ranks.

The log can show LUHMANN-AUTOPOIESES, HEGEL, and Master / Slave(s) group listings. These messages describe coordination, not a new target variable, model family or validation split. Run each independent job in its own working directory and provide the shared files required by the MPI workflow.

C.4 Choosing the time record

Record value	Intended starting use	Enabled speed options
-100 -1 !!	Shorter initial exploration	Faster swap / Cholesky reuse branch and on-the-fly scope selection
-1000 -1 !!	Larger initial exploration budget	The same options with a longer run-through target

Use one of these values for the ordered `TargetCPUTimeBeforeSwap` record. The second field does not impose a positive refinement-time cap, so the whole job can take longer than 100 or 1000 seconds. The dataset, depth schedule, rank count and subsequent Designer interaction analysis also affect the total time.

For Designer post-processing, an additional practical choice is **AnalyzeInteractionLevel 1 1** when detailed interaction chains are not needed. That setting reduces the requested interpretation work; it is separate from the Regressor's three acceleration mechanisms.

End of manual.